

Chapter 2. Chemical Thermodynamics and Transport Properties

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Outlines

1. Chemical Thermodynamics

1.1. Fundamentals of CT

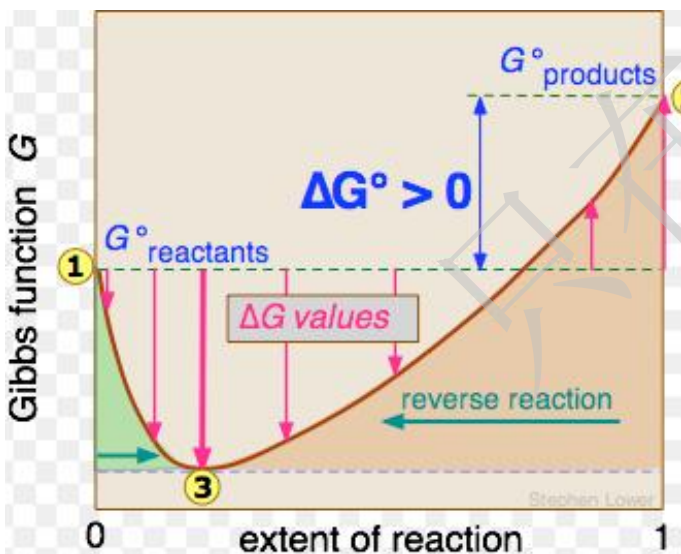
1.2. CT in Combustion Chemistry

2. Transport properties

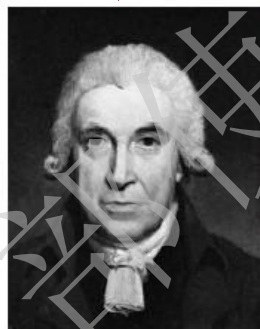
3. Recommended Literature

1. Chemical Thermodynamics

- n Describe the driving force of a chemical process
- n Determine the ideal end state of a reaction process
- n The foundation of chemical kinetics and combustion chemistry



Watt steam engine (1765)



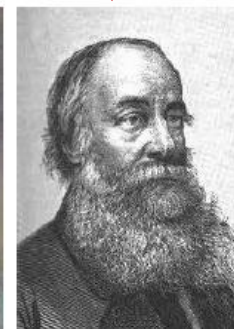
James Watt

Carnot cycle (1824)

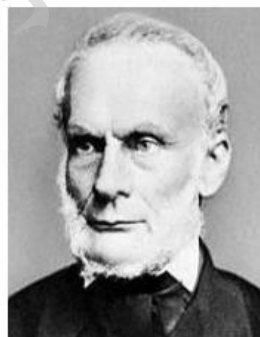


Sadi Carnot

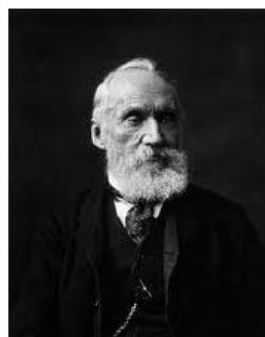
Heat equivalent of work (1840)



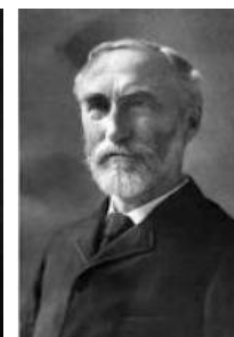
James Prescott Joule



Rudolf J. E. Clausius



William Thomson Kelvin



J Willard Gibbs

Second law of thermodynamics (1850)

Kelvin scale (1848)

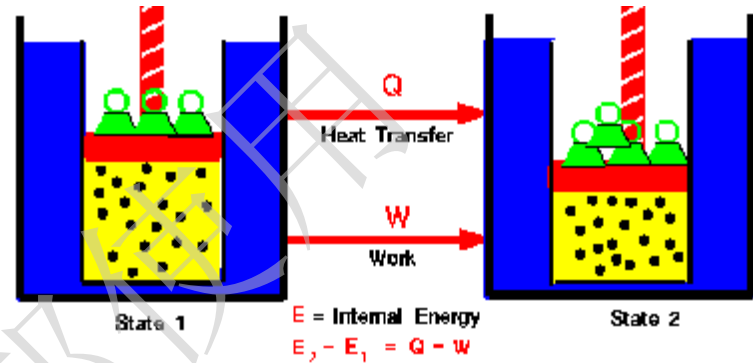
Chemical equilibrium (1876)
Gibbs free energy (1878)

1.1. Fundamentals of CT

I The First Law of Thermodynamics

Conservation of Energy

$$Q - W = \Delta U$$



- n For a working fluid in a closed system, the heat (Q) it receives from the surrounding minus the work (W) it performs on the surrounding is equal to the change in its **internal energy**
- n $\Delta U = U(2) - U(1)$ — $U(2)$: final state; $U(1)$: initial state
- n $Q > 0$: receive heat from the surrounding (**endothermic**)
- n $Q < 0$: release heat to the surrounding (**exothermic**)

1.1. Fundamentals of CT

I The First Law of Thermodynamics

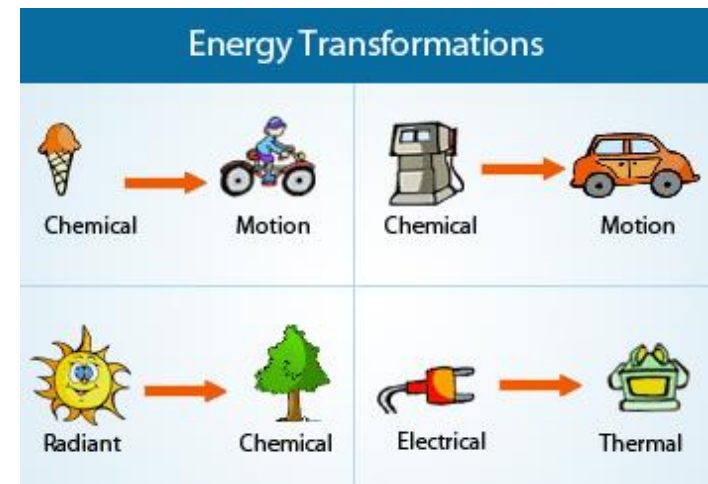
Energy Transformation

n Forms of energy

- p Kinetic energy (motion, thermal ...)
- p Potential energy (gravity, electrical, nuclear...)
- p Chemical energy
- p Radiant energy (light, heat radiation)

n The energy in these forms can be transformed from one to another

n Internal energy is the total energy of molecules in the working fluid - a sum of kinetic and potential energies



1.1. Fundamentals of CT

I Enthalpy

- n If only boundary work is done (e.g., gas expansion)

$$W = P_2V_2 - P_1V_1$$

$$Q = (U_2 + P_2V_2) - (U_1 + P_1V_1)$$



- n Define: **H (enthalpy) = $U + PV$**

Figure 1.15. To create a rabbit out of nothing and place it on the table, the magician must summon up not only the energy U of the rabbit, but also some additional energy, equal to PV , to push the atmosphere out of the way to make room. The total energy required is the enthalpy, $H = U + PV$.

- n Enthalpy: a **state function** that depends only on the prevailing equilibrium state identified by the variables including internal energy, pressure and volume

1.1. Fundamentals of CT

I Heat Capacity

- n For a low-density gas like air or combustion gases, T , P and V are related by the ideal gas law or the equation of state

$$PV = nRT$$

- n The internal energy is primarily a function of T

Constant-volume heat capacity $c_v = \left(\frac{\partial U}{\partial T} \right)_v$

- n The relationship between enthalpy and temperature may be established by defining a constant-pressure heat capacity

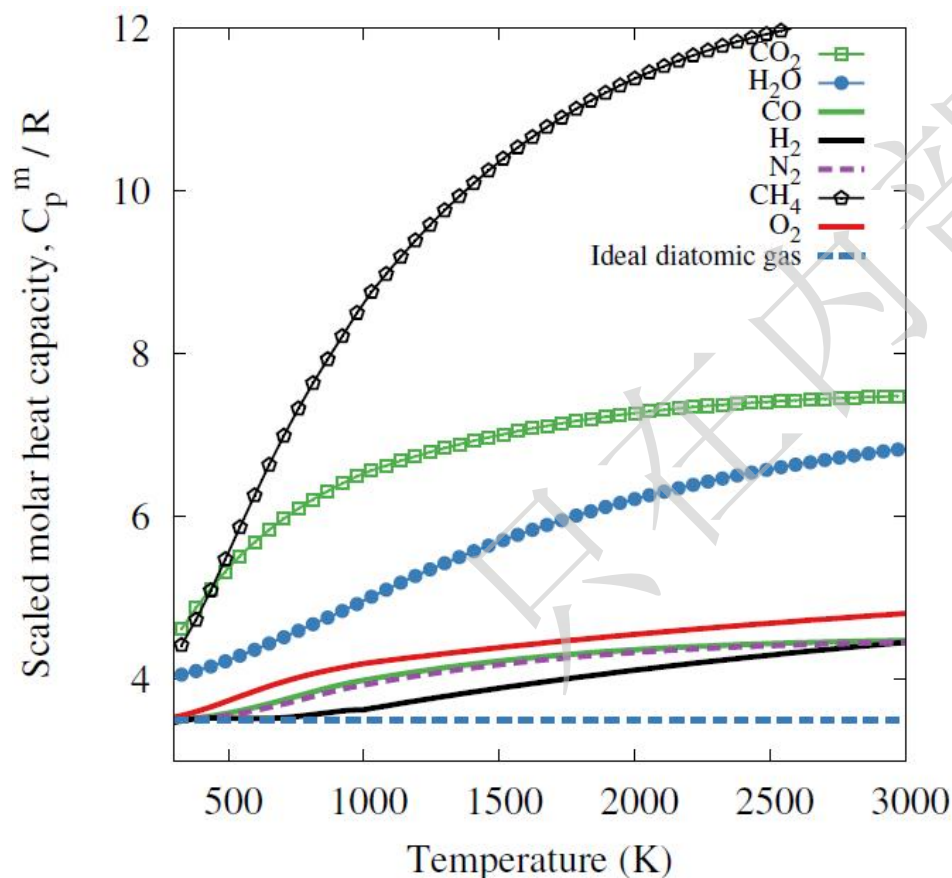
Constant-pressure heat capacity $c_p = \left(\frac{\partial H}{\partial T} \right)_p$

1.1. Fundamentals of CT

I Relationship of Heat Capacity with T

n C_p varies with temperature and species

n The scaled molar heat capacities at constant P ($C_{p,k}^m / R$)



I For non-ideal diatomic or polyatomic gaseous molecules, increasing temperature leads to greater population of excited state and higher freedoms of molecules

I Consequently the $C_{p,k}^m$ becomes higher as the temperature increases

1.1. Fundamentals of CT

I Enthalpy of Formation

Three forms of enthalpy

- n Sensible enthalpy – heating/cooling
- n Latent enthalpy– melting, evaporation/condensation
- n Reaction enthalpy – due to chemical composition change

Enthalpy of formation

- n The heat released from producing 1 mole of a substance from its elements at a constant temperature T
- n The enthalpy of species k is such that:

$$H_k = \int_{T_0}^T C_{p,k} dT + \Delta H_{f,k}^0$$

sensible

mass enthalpy of formation

1.1. Fundamentals of CT

I Enthalpy of Formation

Standard enthalpy of formation of some combustion species

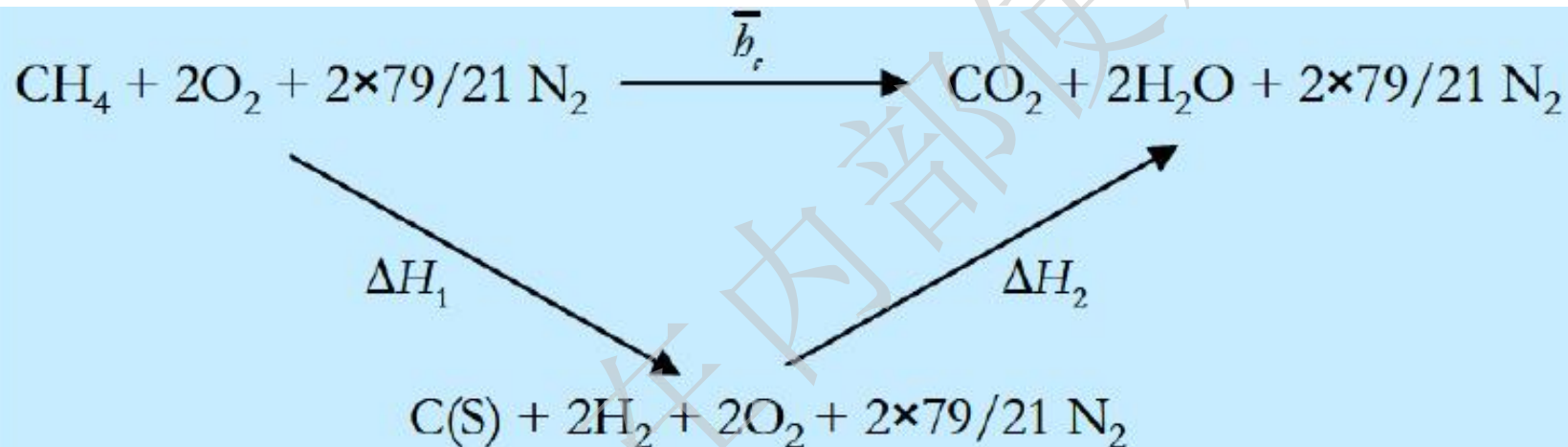
Species	W_k (kg/mol)	$\Delta h_{f,k}^o$ (kJ/kg)	$\Delta h_{f,k}^{o,m}$ (kJ/mole)
CH ₄	0.016	-4675	-74.8
C ₃ H ₈	0.044	-2360	-103.8
C ₈ H ₁₈	0.114	-1829	-208.5
CO ₂	0.044	-8943	-393.5
H ₂ O	0.018	-13435	-241.8
O ₂	0.032	0	0
H ₂	0.002	0	0
N ₂	0.028	0	0

Table: Molecular mass W_k (kg/mol) and formation enthalpies $\Delta h_{f,k}^o$ (kJ/kg), and $\Delta h_{f,k}^{o,m}$ (kJ/mole) at $T_0 = 298.15$ K.

1.1. Fundamentals of CT

I Heat of Combustion

n Definition: Heat released from complete combustion of 1 mole of a fuel



$$b_f (\text{kJ/kmol}) = \Delta H_1 + \Delta H_2 = -1 \times \Delta H_{f298}^0 (\text{CH}_4) + 1 \times \Delta H_{f298}^0 (\text{CO}_2) + 2 \times \Delta H_{f298}^0 (\text{H}_2\text{O})$$

Can also be extended to incomplete combustion system to calculate adiabatic flame temperature (Problem for this chapter)

1.1. Fundamentals of CT

I The Second Law of Thermodynamics

n The Clausius statement

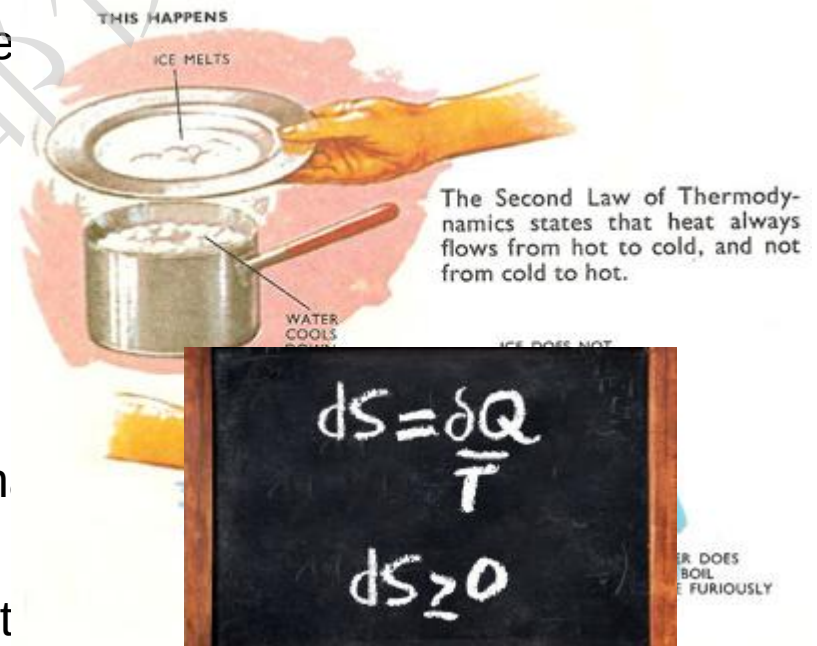
- p Heat can never pass from a colder to a warmer body without some other change, connected therewith, occurring at the same time

n The Kelvin statement

- p It is impossible, by means of inanimate material agency, to derive mechanical effect from any portion of matter by cooling it below the temperature of the coldest of the surrounding objects

n The Kelvin-Planck statement

- p It is impossible to construct a device that will operate in a cycle and produce no effect other than the raising of a weight and the exchange of heat with a single reservoir



Entropy

1.1. Fundamentals of CT

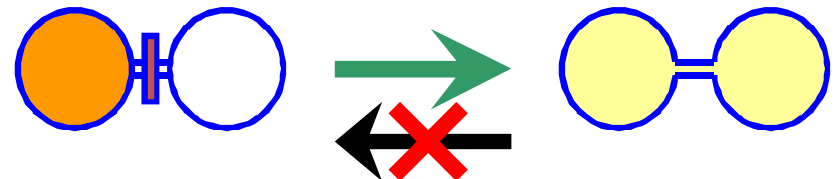
I Entropy

$$S = \left(\frac{\delta Q}{T} \right)_{int rev}$$

$(\delta Q)_{int rev}$: The heat a control mass received during an infinitesimal, internally reversible process

- n From the point of statistical thermodynamics: entropy is a measure of **molecular randomness**
- n For a **spontaneous process** to occur, the entropy of the control mass must be equal to or greater than zero, that is, for a spontaneous process to occur, the entropy of the control volume must increase

$$\Delta S = S_2 - S_1 \geq 0$$



Without heat release or absorption, a spontaneous process will proceed in the direction to increase entropy

1.1. Fundamentals of CT

I The Second Law and Entropy

Entropy — $S(T, P)$

■ Apply the first law to a reversible constant T and P process

$$\delta Q_{int rev} - \delta W_{int rev} = dU \quad \begin{aligned} \delta Q_{int rev} &= TdS \\ \delta W_{int rev} &= PdV \end{aligned}$$

$$TdS = dU + PdV \quad \downarrow \quad dS = \frac{dU}{T} + P\frac{dV}{T}$$

$$dS = C_v \frac{dT}{T} + R \frac{dV}{V} \quad \downarrow \quad dS = C_p \frac{dT}{T} - R \frac{dP}{P}$$

$$\Delta S = S_2 - S_1 = \int_1^2 \left(C_v \frac{dT}{T} \right) + R \ln \frac{V_2}{V_1} \quad \Delta S = S_2 - S_1 = \int_1^2 \left(C_p \frac{dT}{T} \right) - R \ln \frac{P_2}{P_1}$$

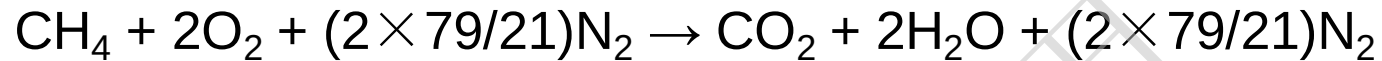
■ Standard-state entropy (1 atm): $\Delta S = \int_0^T \left(C_p \frac{dT}{T} \right)$

S increases as T and V increases, but decreases with P increases

Third law: $S = 0$ at 0 K

1.1. Fundamentals of CT

I Chemical Equilibrium



Complete combustion, greater heat release



Incomplete combustion, less heat release, more entropy

- n How does the reaction know when to stop?
- n In practical combustion processes, complete combustion can never happen. Instead the combustion will reach the state of **chemical equilibrium**
- n A compromise between enthalpy and entropy releases must be made!

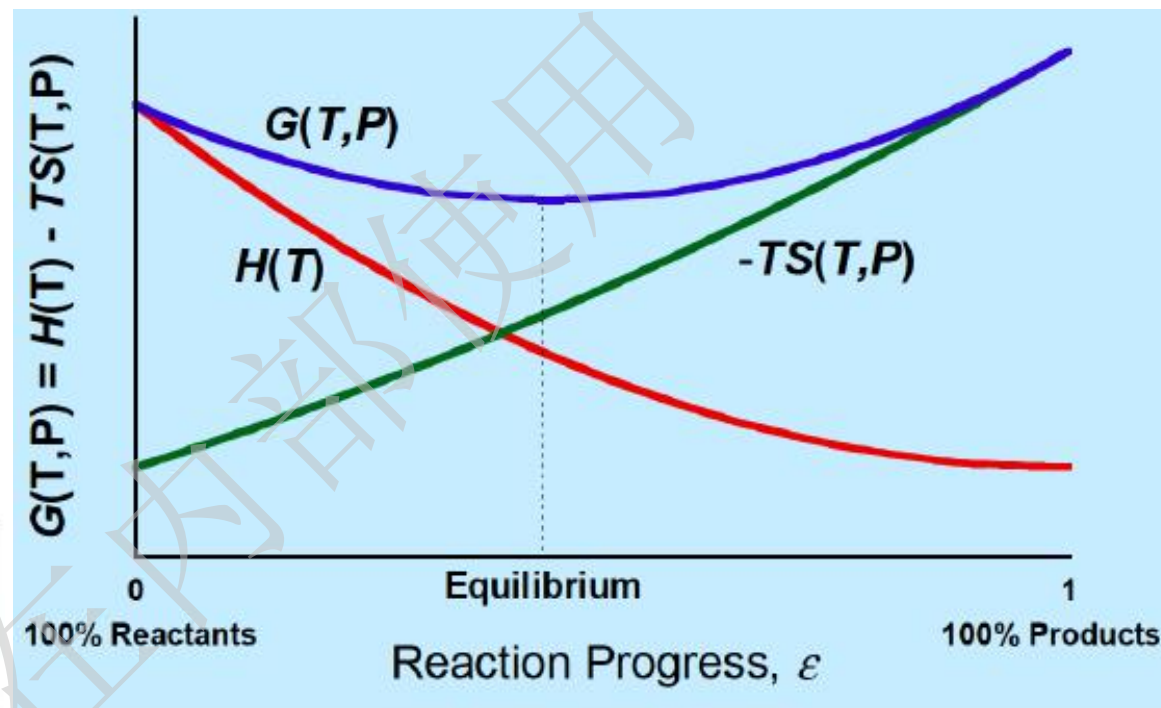
1.1. Fundamentals of CT

I Gibbs Free Energy

$$G = H - TS$$

$$\Delta G = \Delta H - T\Delta S$$

	$\Delta H < 0$	$\Delta H > 0$
$\Delta S > 0$	Spontaneous	Spontaneous at high temps
$\Delta S < 0$	Spontaneous at low temps	Not Spontaneous

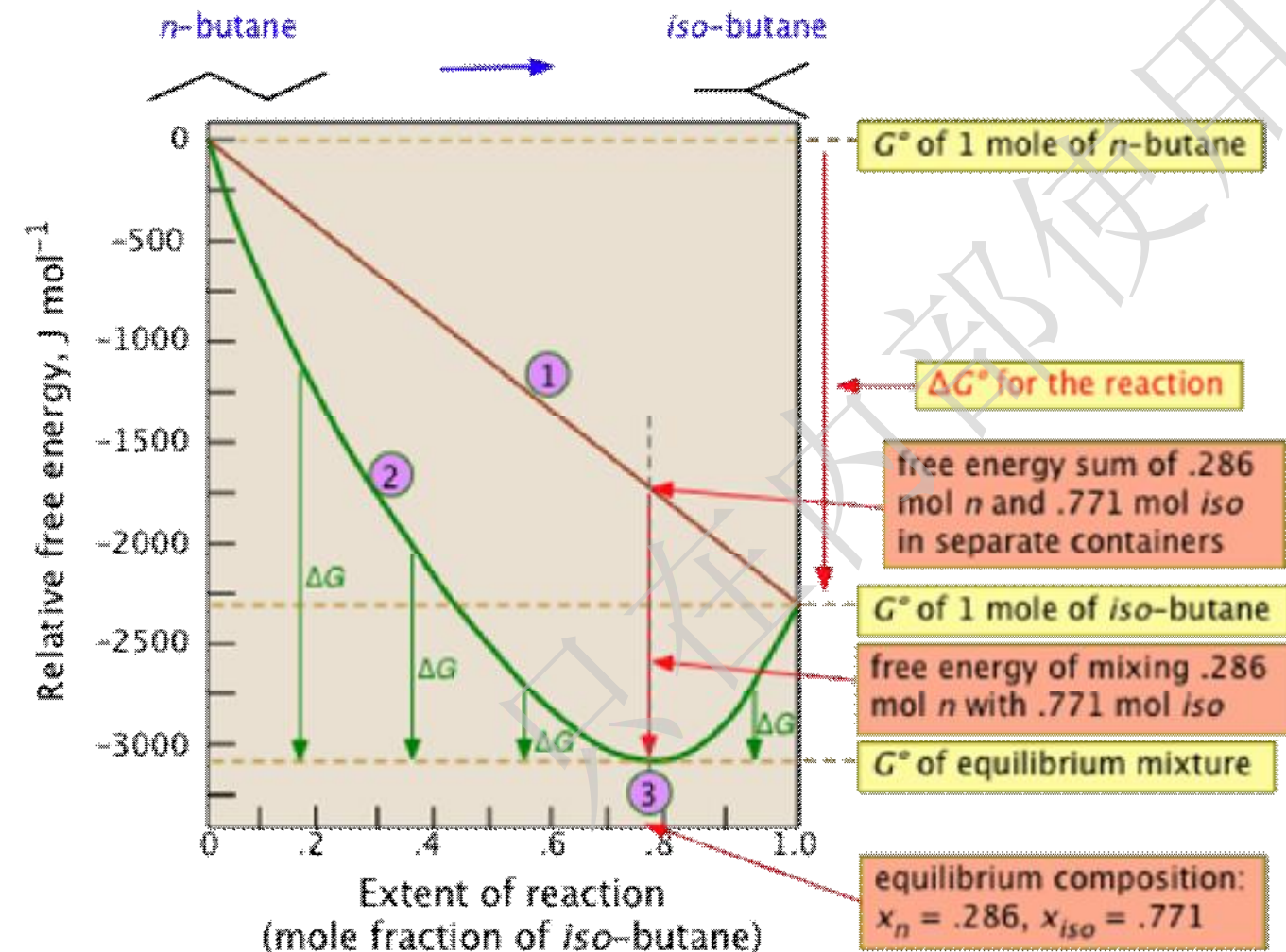


n The free enthalpy G (Gibbs energy) indicates if a reaction is **spontaneously possible** ($\Delta G < 0$) or not

n Equilibrium is found when $dG/d\varepsilon = 0$

1.1. Fundamentals of CT

I Gibbs Free Energy - Example



- n The equilibrium composition corresponding to ~77 % conversion
- n G determines the extend of a reaction processes

1.1. Fundamentals of CT

I Equilibrium Constant

Reaction $\sum_{\text{react.}} v_i A_i \rightarrow \sum_{\text{prod.}} v_{i'} A_{i'}$

Gibbs free energy $G(T, P) = \sum_{\text{react.}} n_i G_i(T, P) + \sum_{\text{prod.}} n_{i'} G_{i'}(T, P)$

Equilibrium at const. T & P $\frac{dG(T, P)}{d\varepsilon} = 0 = \sum_{\text{react.}} g_i(T, P) \frac{dn_i}{d\varepsilon} + \sum_{\text{prod.}} g_{i'}(T, P) \frac{dn_{i'}}{d\varepsilon}$

Apply mass conservation $-\sum_{\text{react.}} v_i g_i(T, P) + \sum_{\text{prod.}} v_{i'} g_{i'}(T, P) = 0$

Gibbs function of species i $g_i(T, P) = g_i^0(T) + RT \ln\left(\frac{P}{P_0}\right)$

$\Delta G_r^0(T) = -R_u T \ln \left(\frac{\prod_{\text{prod.}} P_{i'}^{v_{i'}}}{\prod_{\text{react.}} P_i^{v_i}} \right)$

$K_p(T) \equiv \frac{\prod_{\text{prod.}} P_{i'}^{v_{i'}}}{\prod_{\text{react.}} P_i^{v_i}} = \exp \left[-\frac{\Delta G_r^0(T)}{R_u T} \right]$

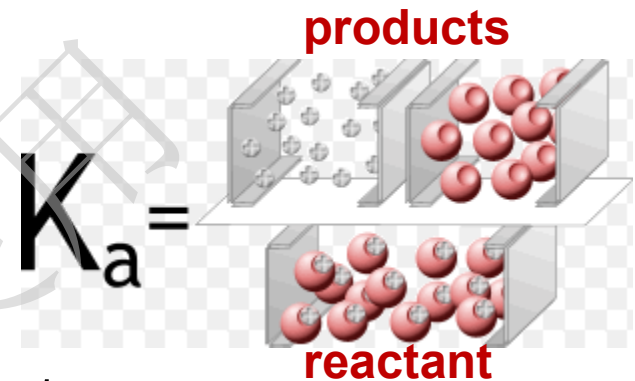
Equilibrium Constant

1.1. Fundamentals of CT

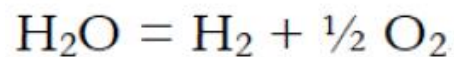
I Equilibrium Constant

n K_p is a function of temperature only

$$K_p(T) \equiv \frac{\prod_{\text{prod.}} P_i^{v_i}}{\prod_{\text{react.}} P_i^{v_i}} = \exp \left[-\frac{\Delta G_r^\circ(T)}{R_u T} \right]$$



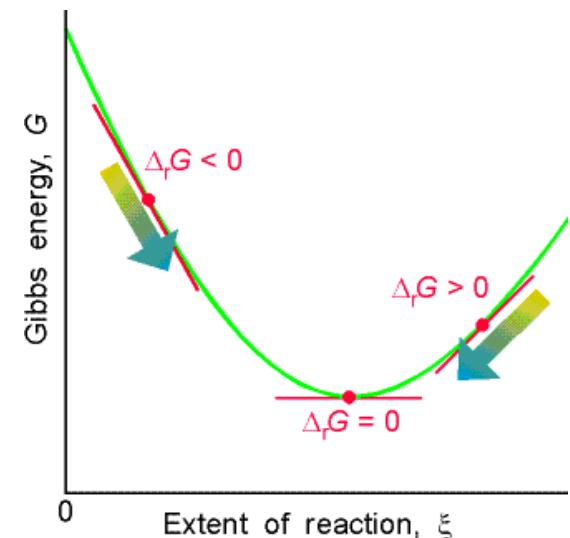
n Forward/backward equilibrium constant



$$K_{p,f}(T) = \frac{P_{\text{H}_2} P_{\text{O}_2}^{1/2}}{P_{\text{H}_2\text{O}}}$$

$$K_{p,b}(T) = \frac{P_{\text{H}_2\text{O}}}{P_{\text{H}_2} P_{\text{O}_2}^{1/2}}$$

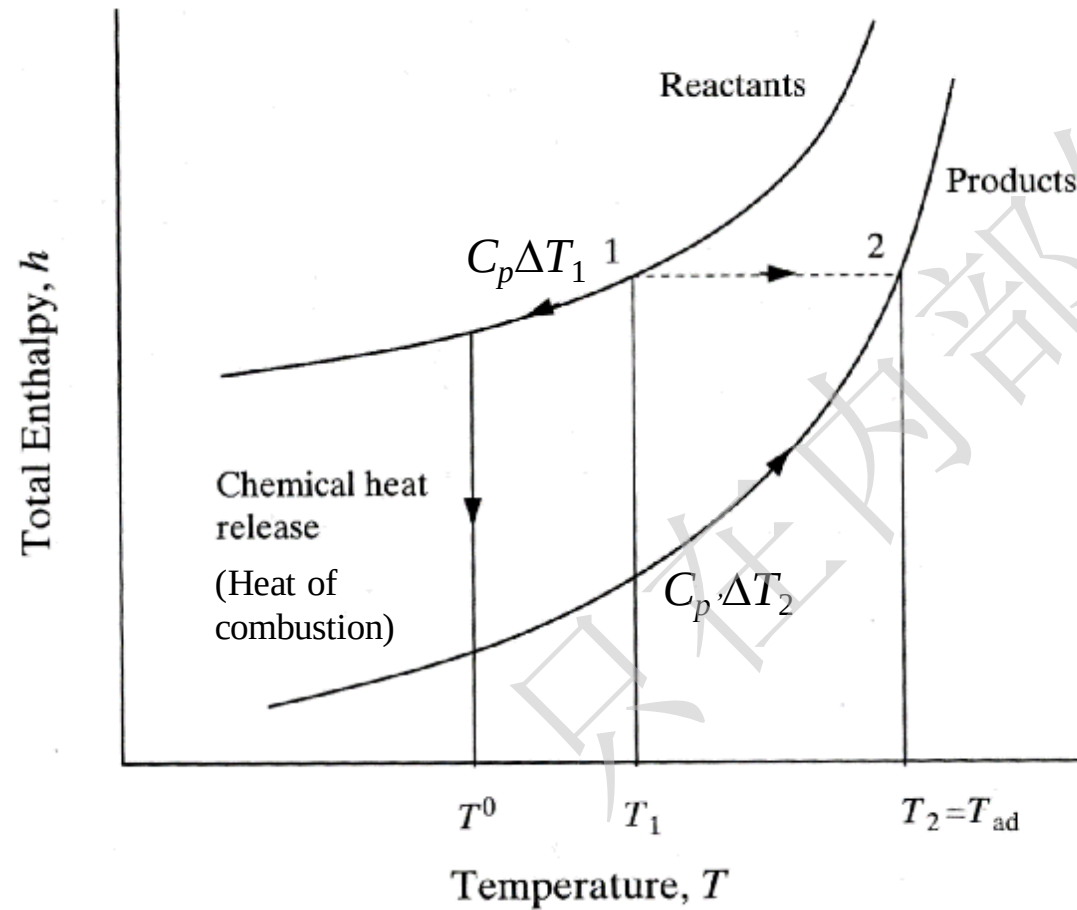
$$K_{p,f}(T) = \frac{1}{K_{p,b}(T)}$$



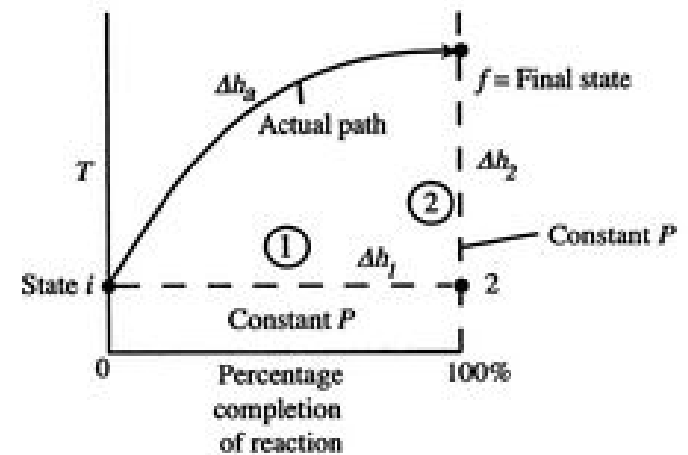
n The larger the equilibrium constant, the greater the completion of a reaction

1.1. Fundamentals of CT

I Adiabatic Flame Temperature



- n The adiabatic flame temperature is obtained when all the heat released from combustion reaction is used to raise the product temperature from T^0 (usually 298 K) to T_{ad}

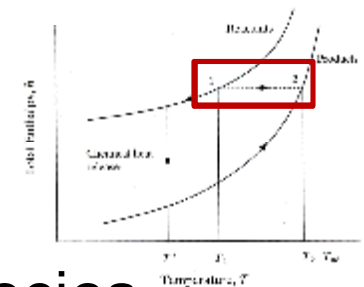


1.1. Fundamentals of CT

I Adiabatic Flame Temperature



n Adiabaticity gives $H_{\text{prod.}}(T_{ad}) - H_{\text{react.}}(T_1) = 0$



n Using the total enthalpy equation for each species

$$\sum_{\text{prod.}} \nu_i \bar{h}_{T_{ad},i} - \sum_{\text{react.}} \nu_i \bar{h}_{T_0,i} = \left[\sum_{\text{prod.}} \nu_i \bar{h}_{f,298,i}^0 - \sum_{\text{react.}} \nu_i \bar{h}_{f,298,i}^0 \right] + \sum_{\text{prod.}} \nu_i [\bar{h}(T_{ad}) - \bar{h}(298)]_i - \sum_{\text{react.}} \nu_i [\bar{h}(T_1) - \bar{h}(298)]_i = 0$$

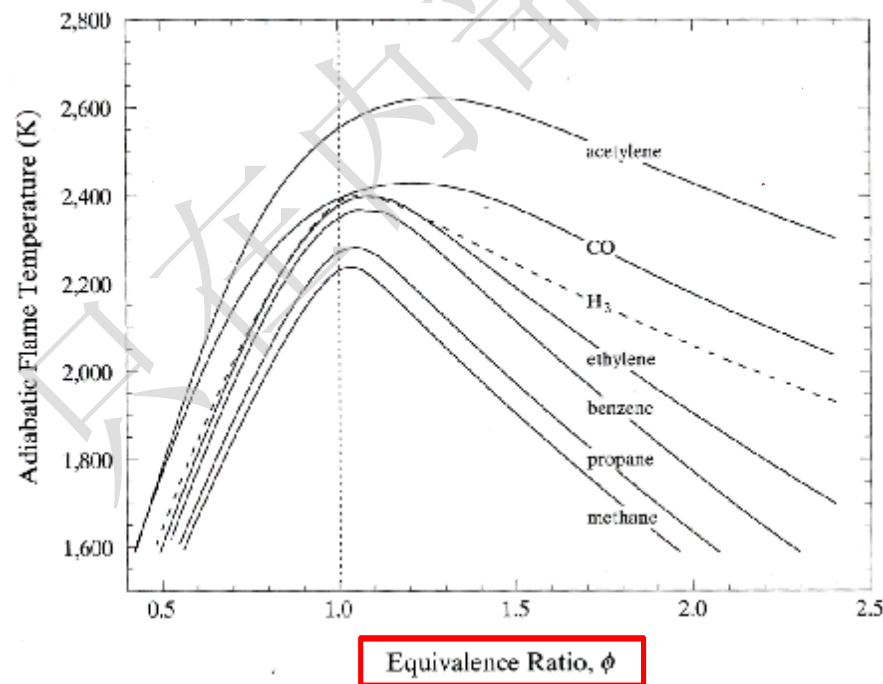
n Let $T_1 = 298 \text{ K}$

$$-\Delta H_{r,298}^{\circ} = \sum_{\text{prod.}} \nu_i [\bar{h}(T_{ad}) - \bar{h}(298)]_i$$

1.1. Fundamentals of CT

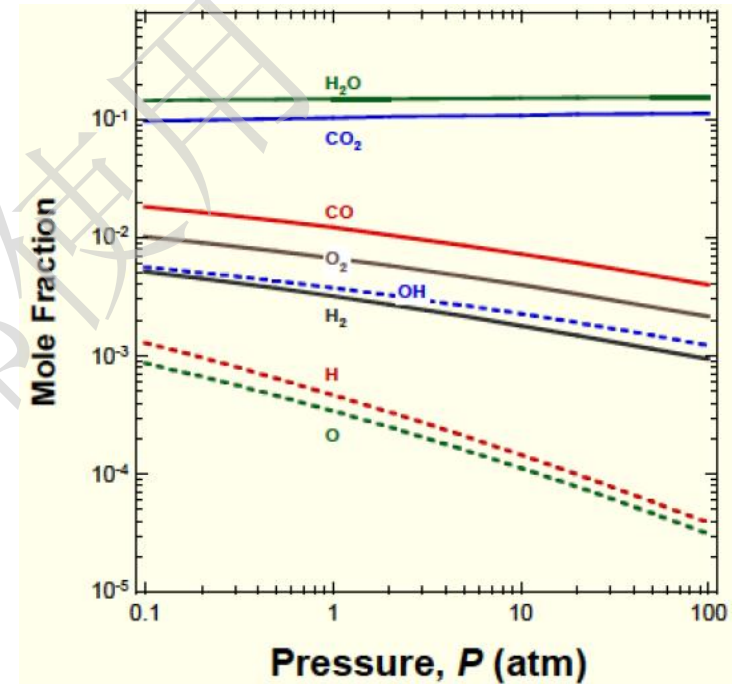
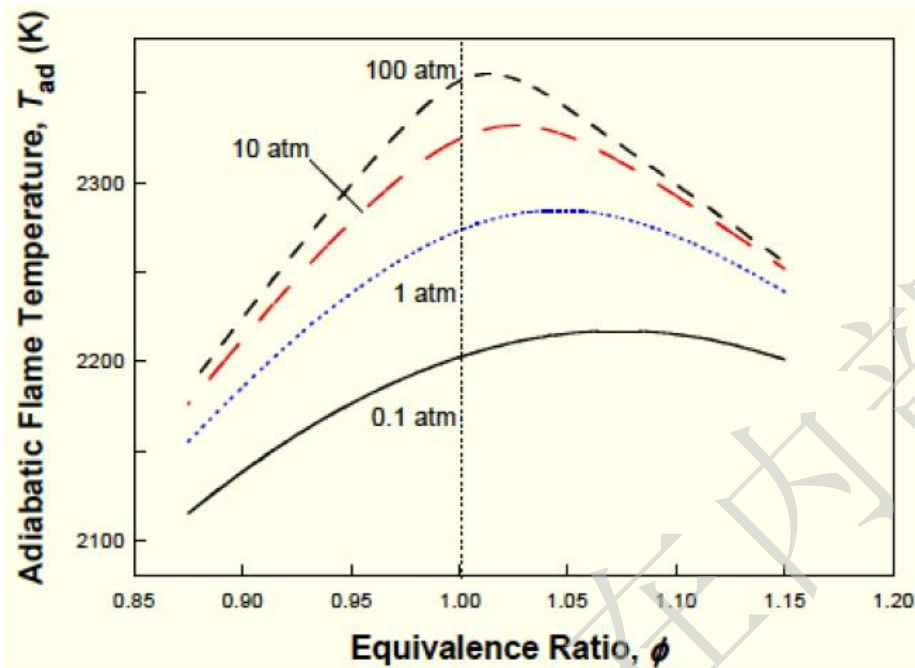
I Adiabatic Flame Temperature

- n The decrease of T_{ad} towards smaller ϕ is caused by dilution of unused oxygen as well as the greater amount of nitrogen brought into the combustor with oxygen.
- n The decrease of T_{ad} towards larger ϕ is because of oxygen deficiency, which does not allow the fuel to be fully oxidized.



1.1. Fundamentals of CT

I Adiabatic Flame Temperature

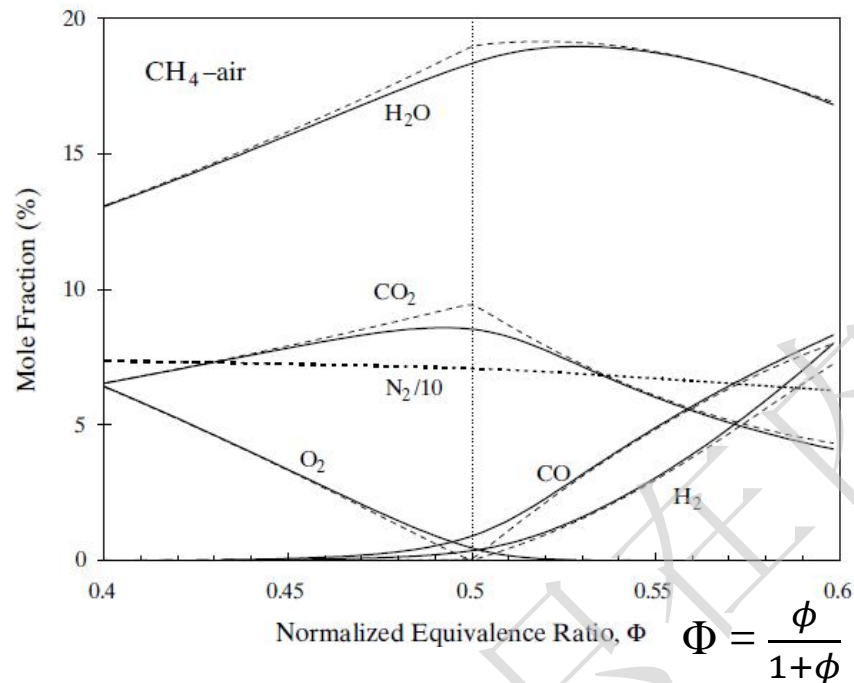


- n As pressure increases, (a) the adiabatic flame temperature becomes higher and (b) the peak shifts towards $\phi = 1$.
- n The pressure serves to reduce the extent of dissociation of CO_2 and H_2O , and in doing so, forces the reaction towards greater completion.

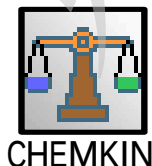
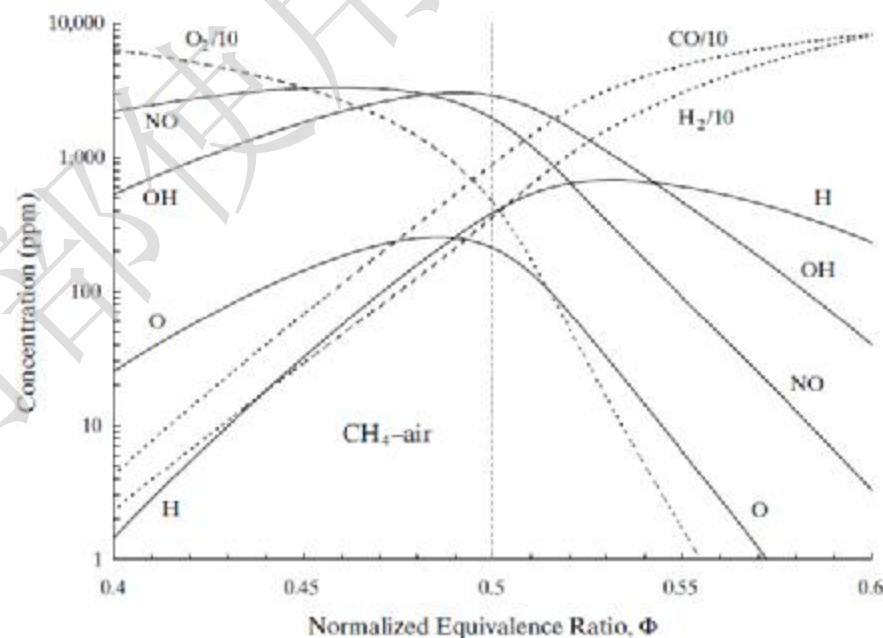
1.1. Fundamentals of CT

I Equilibrium Composition – CH₄/air Flame

Major flame species



Minor flame species



www.gaseq.co.uk



www.grc.nasa.gov/WWW/CEAWeb

A lot of commercial or online calculation programs

1.2. CT in Combustion Chemistry

I Thermodynamics Data

Parameters needed in kinetic modeling

n Thermochemical data for all species

p Enthalpy of formation ΔH_f^0

p Entropy S^0

p Heat capacity C_p^0

Necessary for the calculation of

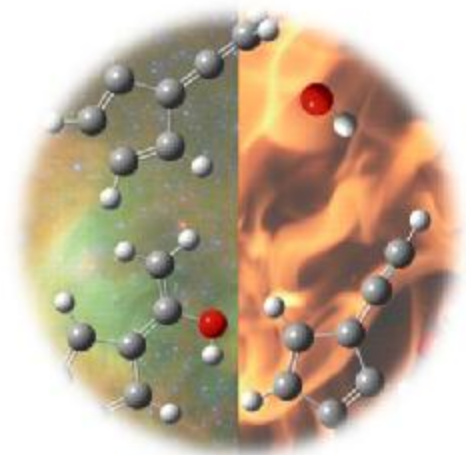
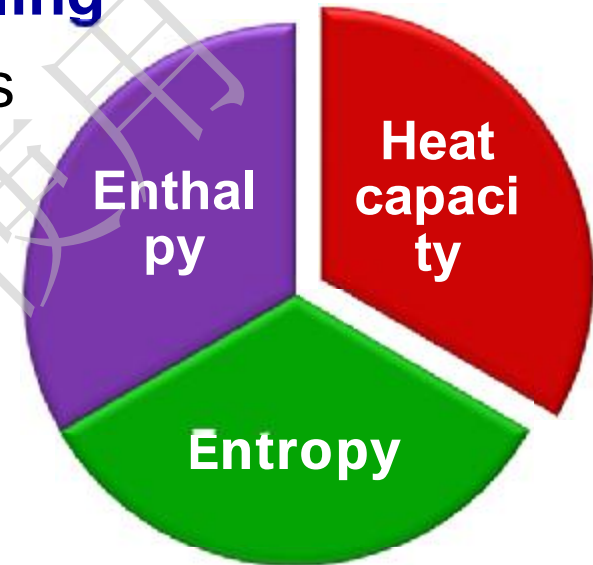
n Thermal effect of the reaction

n Energy conversion equation

n Equilibrium constant of the reaction

n Equilibrium composition and T_{ad}

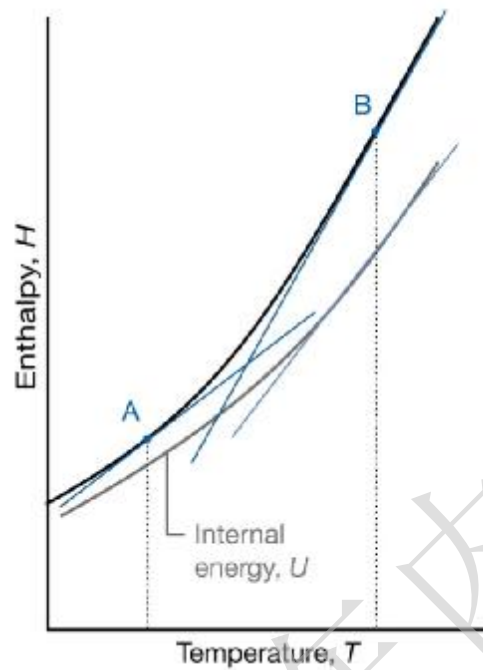
n Relationship between direct and forward rate constant of an elementary reaction



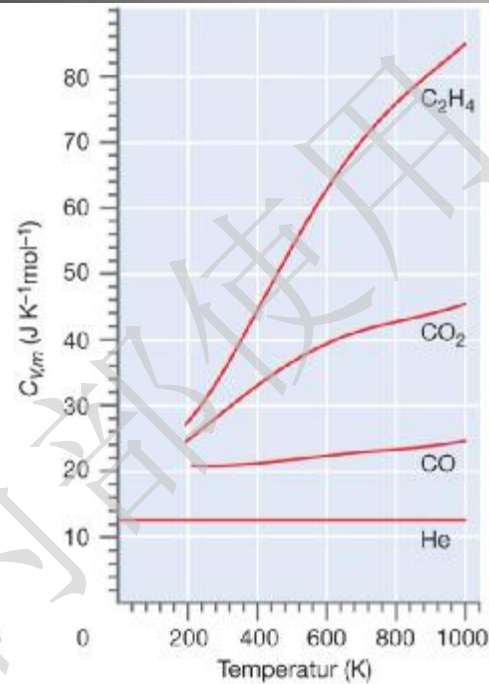
1.2. CT in Combustion Chemistry

I Temperature Dependence

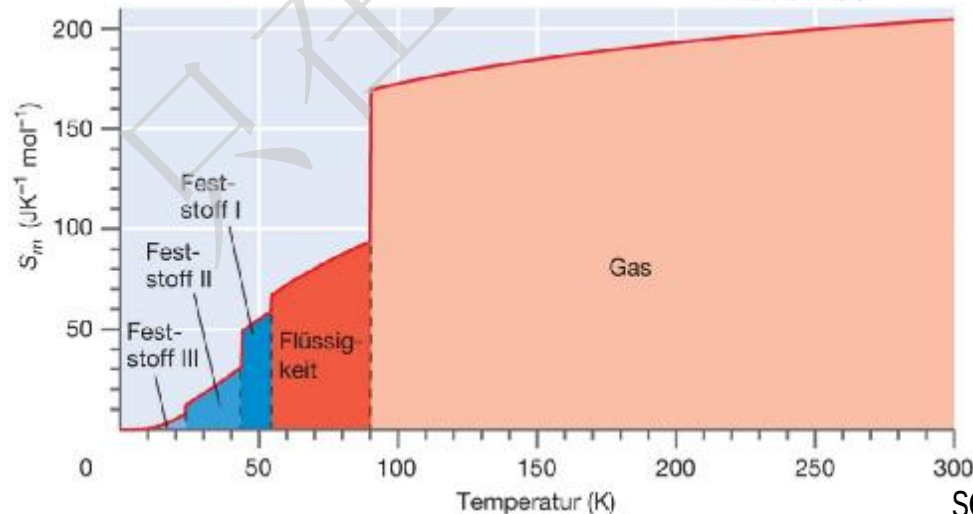
Enthalpy



Heat capacity



Entropy



1.2. CT in Combustion Chemistry

I NASA Polynomials

Species name	Reference source	Composition	Phase	T_{Low}	T_{high}	T_{mid}	
CO2	L 7/88C	10 2 0 0G	200.000	6000.000	1000.		1
$a_i (i = 1,7) \text{ for } T_{mid} < T < T_{high}$							2
$a_i (i = 1,7) \text{ for } T_{low} < T < T_{mid}$							3
							4

$$\bar{c}_p / R_u = a_1 + a_2 T + a_3 T^2 + a_4 T^3 + a_5 T^4,$$

$$\bar{s}^\circ / R_u = a_1 \ln T + a_2 T + a_3 T^2 / 2 + a_4 T^3 / 3 + a_5 T^4 / 4 + a_7.$$

$$\bar{h}_T / R_u = a_1 T + a_2 T^2 / 2 + a_3 T^3 / 3 + a_4 T^4 / 4 + a_5 T^5 / 5 + a_6$$

$$\bar{h}(T) - \bar{h}(298.15) = \bar{h}_T(T) - \bar{h}_T(298.15)$$

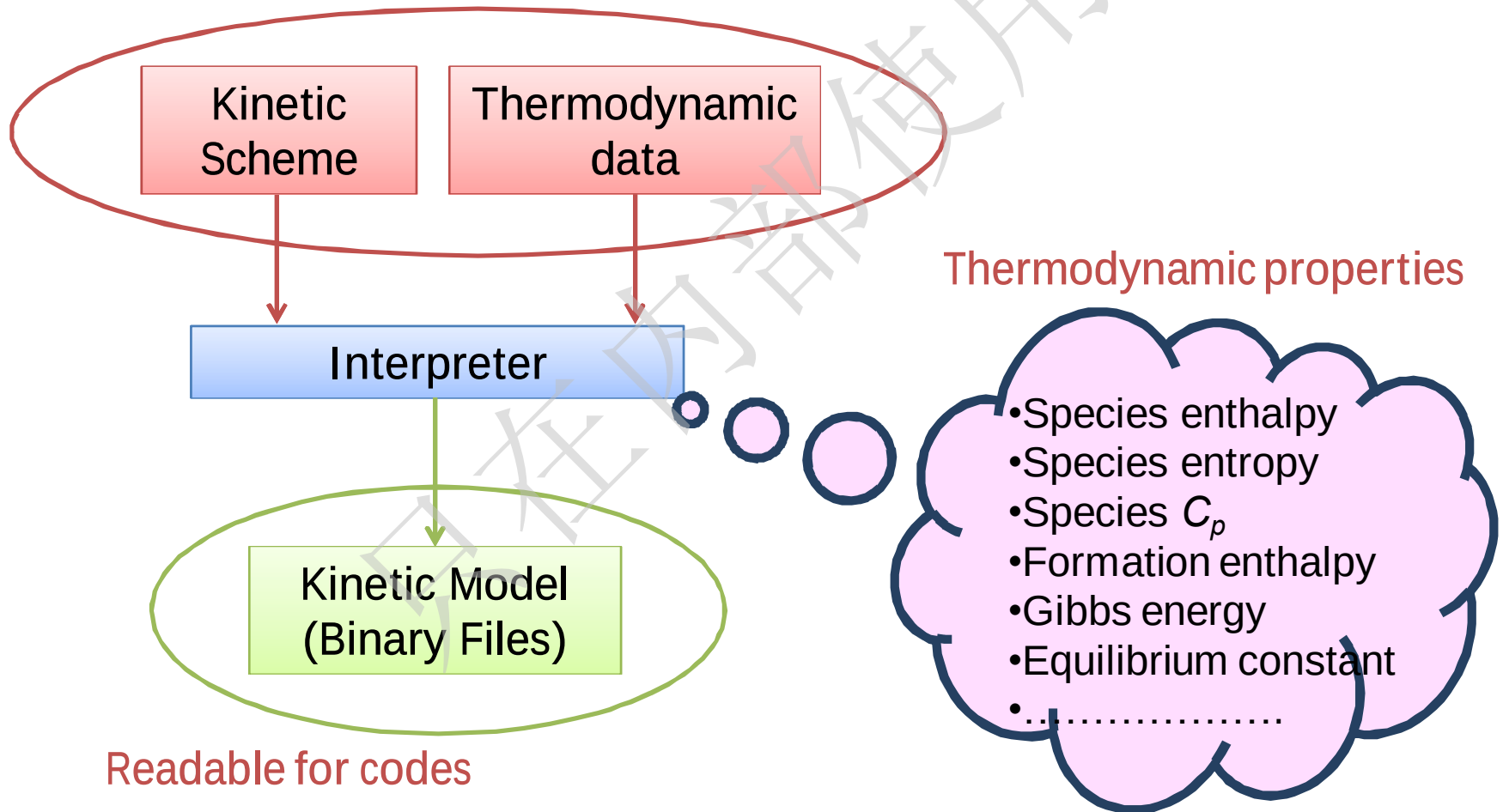
$$\bar{h}_f^\circ(T) = \bar{h}_T(T) - \sum_{\text{elements}} \nu_i \bar{h}_{T,i}(T)$$

- Adopted in many databases
- Used in almost all computational codes

1.2. CT in Combustion Chemistry

I Interpretation of Thermodynamic Data

CHEMKIN/CANTERA/OPENSMOKE/FLAMEMASTER/COSILAB/LOGE/FLUENT.....



1.2. CT in Combustion Chemistry

I Source of Thermodynamic Data

Online databases

NIST-JANAF Thermochemical Tables

n <http://webbook.nist.gov/chemistry/>

n <http://kinetics.nist.gov/janaf/>

n <http://atct.anl.gov/>

n <http://garfield.chem.elte.hu/burcat/burcat.html>

Methane (CH ₄)					C ₁ H ₄ (g)		
Enthalpy Reference Temperature = $T_r = 298.15 \text{ K}$					Standard State Pressure = $p^\circ = 0.1 \text{ MPa}$		
T/K	C_p°	S°	$-[G^\circ - H(T_r)]/T$	$H - H(T_r)$	$\Delta_f H^\circ$	$\Delta_f G^\circ$	$\log K_f$
0	0.	0.	INFINITE	-10.024	-66.911	-66.911	INFINITE
100	33.258	149.500	216.485	-6.698	-69.644	-64.353	33.615
200	33.473	172.577	189.418	-3.368	-72.027	-58.161	15.190
250	34.216	180.113	186.829	-1.679	-73.426	-54.536	11.395
298.15	35.639	186.251	186.251	0.	-74.873	-50.768	8.894
300	35.708	186.472	186.252	0.066	-74.929	-50.618	8.813
350	37.874	192.131	186.694	1.903	-76.461	-46.445	6.932
400	40.500	197.356	187.704	3.861	-77.969	-42.054	5.492
450	43.374	202.291	189.053	5.957	-79.422	-37.476	4.350
500	46.342	207.014	190.614	8.200	-80.802	-32.741	3.420
600	52.227	215.987	194.103	13.130	-83.308	-22.887	1.993
700	57.794	224.461	197.840	18.635	-85.452	-12.643	0.943
800	62.932	232.518	201.675	24.675	-87.238	-2.115	0.138
900	67.601	240.205	205.532	31.205	-88.692	8.616	-0.500
1000	71.795	247.549	209.370	38.179	-89.849	19.492	-1.018

1.2. CT in Combustion Chemistry

I Source of Thermodynamic Data

Research papers



Critical Evaluation of Thermochemical Properties of C1–C4 Species: Updated Group-Contributions to Estimate Thermochemical Properties

S. M. Burke, J. M. Simmie, and H. J. Curran

Citation: **Journal of Physical and Chemical Reference Data** 44, 013101 (2015) doi: 10.1063/1.4902535

View online: <http://dx.doi.org/10.1063/1.4902535>

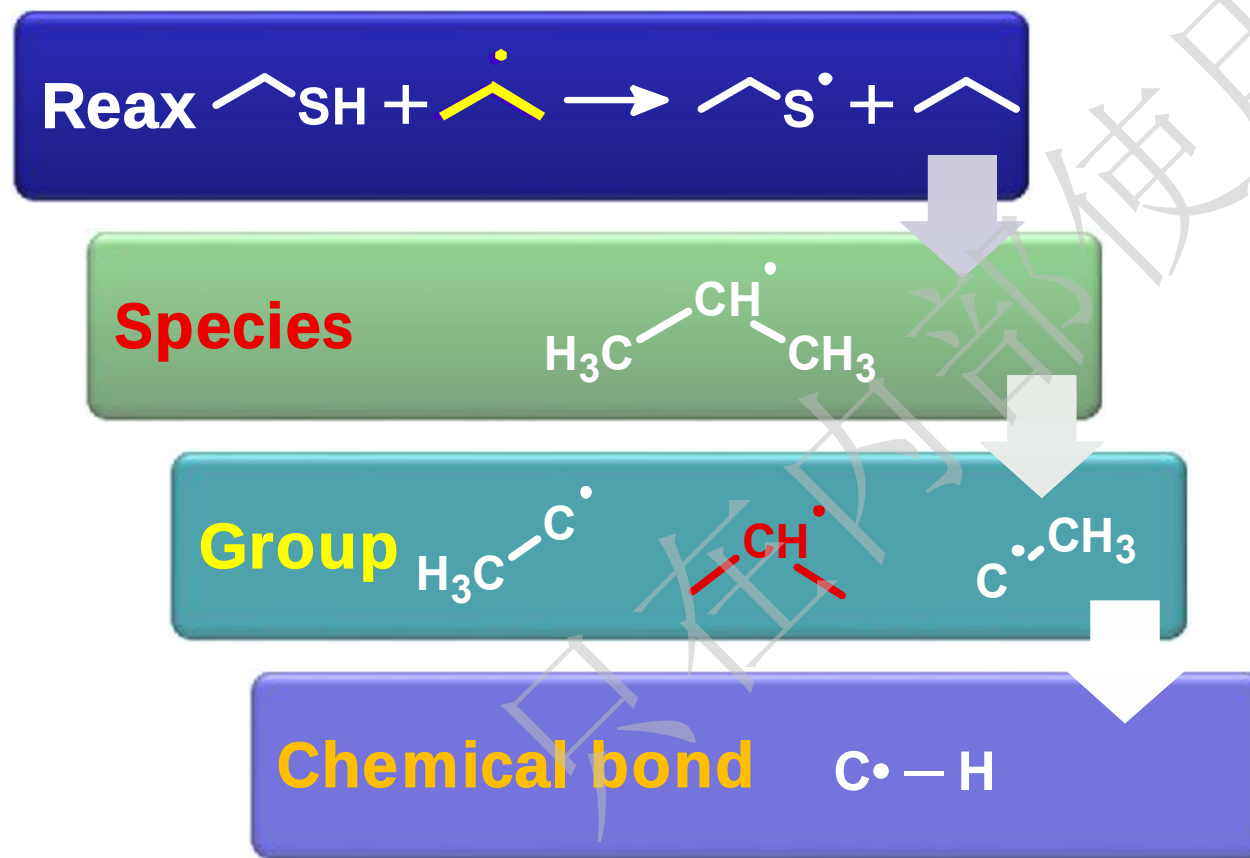
View Table of Contents: <http://scitation.aip.org/content/aip/journal/jpcrd/44/1?ver=pdfcov>

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1.2. CT in Combustion Chemistry

I Source of Thermodynamic Data

Group Additivity Method



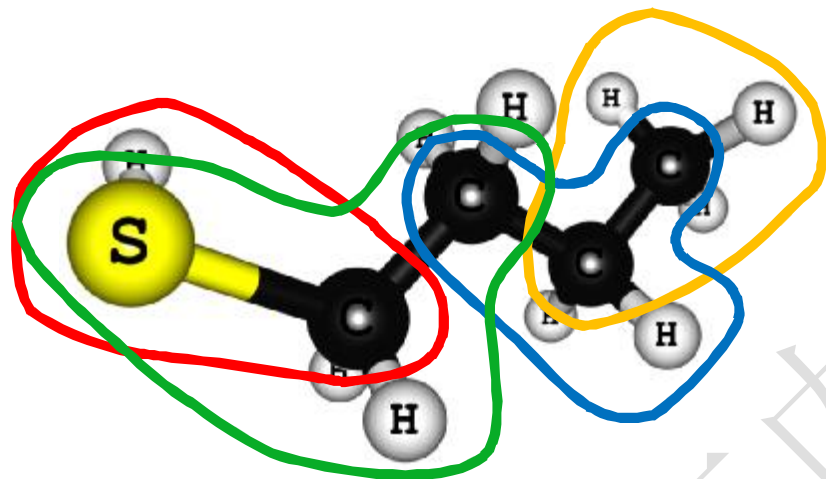
$$K_r = e^{-\frac{D_r G^\circ}{RT}}$$
$$D_r G^\circ = \sum_i v_i D_f G^\circ(i)$$
$$D_f G^\circ(i) = \sum_g D_{gr} G^\circ(g)$$
$$D_{gr} G^\circ(g) = \sum_b D_{bond} G^\circ(b)$$

- n The thermochemical properties of a species are additive with respect to the functional groups that makes up the molecule

1.2. CT in Combustion Chemistry

I Source of Thermodynamic Data

Benson's Group Additivity Method



n Cutting of molecules into groups

Groups: 1) Atom with a valence ≥ 2 ; 2) Atoms or groups of monovalent atoms bonded to it.

Example: C-(C)(H)₃

C-(C)₂(H)₂

S-(C)(H)

C-(S)(C)(H)₂

+ Correction between non-neighboring atoms

$$\Delta_f H^\circ(298 \text{ K}) = \sum_i GAV_{\Delta_f H^\circ}(X_i)$$

$$S^\circ(298 \text{ K}) = \sum_i GAV_{S^\circ}(X_i) - R \ln(\sigma/n_{opt})$$

$$C_p^\circ(T) = \sum_i GAV_{C_p^\circ}(X_i)$$

1.2. CT in Combustion Chemistry

I Source of Thermodynamic Data

Benson's Group Additivity Method

Groups	$\bar{h}_{f,298}^{\circ}$	$\bar{s}_{298}^{\circ}$	$\bar{c}_p(T)$ (J/mol)						
	(kJ/mol)	(J/mol-K)	300	400	500	600	800	1000	1500
C-(C,H ₃)	-42.68	127.24	25.90	32.80	39.33	45.15	54.48	61.80	73.55
C-(C _b ,H ₃)	-42.68	127.24	25.90	32.80	39.33	45.15	54.48	61.80	73.55
C-(C _d ,H ₃)	-42.68	127.24	25.90	32.80	39.33	45.15	54.48	61.80	73.55
C-(C _t ,H ₃)	-42.68	127.24	25.90	32.80	39.33	45.15	54.48	61.80	73.55
C-(C ₂ ,H ₂)	-20.63	39.41	23.01	29.08	34.52	39.12	46.32	51.63	59.41
C-(C ₃ ,H)	-7.95	-50.50	19.00	25.10	30.00	33.68	38.95	42.05	46.78
C-(C ₄)	2.09	-146.86	18.28	25.65	30.79	33.97	36.69	36.65	33.97
C _d -(H ₂)	26.19	115.52	21.34	26.61	31.42	35.56	42.13	47.15	55.19
C _d -(C,H)	35.94	33.35	17.41	21.05	24.31	27.20	32.01	35.35	40.25
C _d -(C ₂)	43.26	-53.14	17.15	19.29	20.88	22.01	24.27	25.44	26.61
C _d -(C _d ,H)	28.37	26.69	18.66	24.23	28.24	31.05	34.94	38.12	42.22
C _d -(C,C _d)	37.15	-61.09	18.41	22.47	24.81	25.86	27.20	27.70	28.12
C _d -(C _b ,H)	28.37	26.69	18.66	24.23	28.24	31.05	34.94	38.12	42.22
C _d -(C,C _b)	36.15	-61.09	18.41	22.47	24.81	25.86	27.20	27.70	28.12
C _d -(C _b ,C _d)	30.04	-69.04	19.66	25.65	28.74	29.71	30.12	29.96	29.54
C _d -(C _d ,C _t)	29.04	-69.04	19.66	25.65	28.74	29.71	30.12	29.96	29.54
C _d -(C _b ,C _t)	28.03	-69.04	19.66	25.65	28.74	29.71	30.12	29.96	29.54
C _d -(C _{d2})	19.25	-69.04	19.66	25.65	28.74	29.71	30.12	29.96	29.54
C _d -(C _{t2})	27.03	-69.04	19.66	25.65	28.74	29.71	30.12	29.96	29.54
C _d -(C _{b2})	33.47	-69.04	19.66	25.65	28.74	29.71	30.12	29.96	29.54
C _d -(C _t ,H)	28.37	33.60	18.66	24.23	28.24	31.05	34.94	38.12	42.22
C _d -(C _b ,C _t)	35.69	61.09	18.41	22.47	24.81	25.86	27.20	27.70	28.12

1.2. CT in Combustion Chemistry

I Source of Thermodynamic Data

Benson's Group Additivity Method

Uncertainties

- Benson's **original** data: ± 2 kcal/mol for molecules, ± 4 kcal/mol for radicals

New groups:

- Domalki and Hearing (1993)
- Cohen (1995)
- W.H. Green (2002)
- Bozzelli and Curran in THERM code

New groups, new corrections and the accurate theoretical calculated group values make the uncertainties smaller!

Codes based on
Benson's group
additivity

& Thergas
& THERM
& CanTherm
& CHETAH
&

1.2. CT in Combustion Chemistry

I Source of Thermodynamic Data

Statistical Thermodynamics

n Partition function measures the total number of possible energy states a molecule can be in at an equilibrium temperature

Total partition function

$$Q = Q_{trans} Q_{rot} Q_{vib} Q_{elec}$$

Translational partition function

$$Q_{trans} = \left(\frac{2\pi M k_B T}{h^2} \right)^{\frac{3}{2}} V$$

Rotational partition function

$$Q_{rot} = \frac{1}{\sigma} \left(\frac{8\pi^2 k_B T}{h^2} \right)^{\frac{3}{2}} \sqrt{I_a I_b I_c}$$

Vibrational partition function

$$Q_{vib} = \prod_i e^{-h\nu_i/2kT} \cdot (1 - e^{-h\nu_i/k_B T})^{-1}$$

Electronic partition function

$$Q_{elec} = \sum_{i=1} g_i \exp\left(-\frac{E_i}{k_B T}\right)$$

1.2. CT in Combustion Chemistry

I Source of Thermodynamic Data

Quantum Chemistry Methods

Enthalpy and heat capability

- n Total partition function to calculate enthalpy

$$S = Nk_B \ln Q + Nk_B T \frac{d \ln Q}{dT}$$

- n Total partition function to calculate internal energy

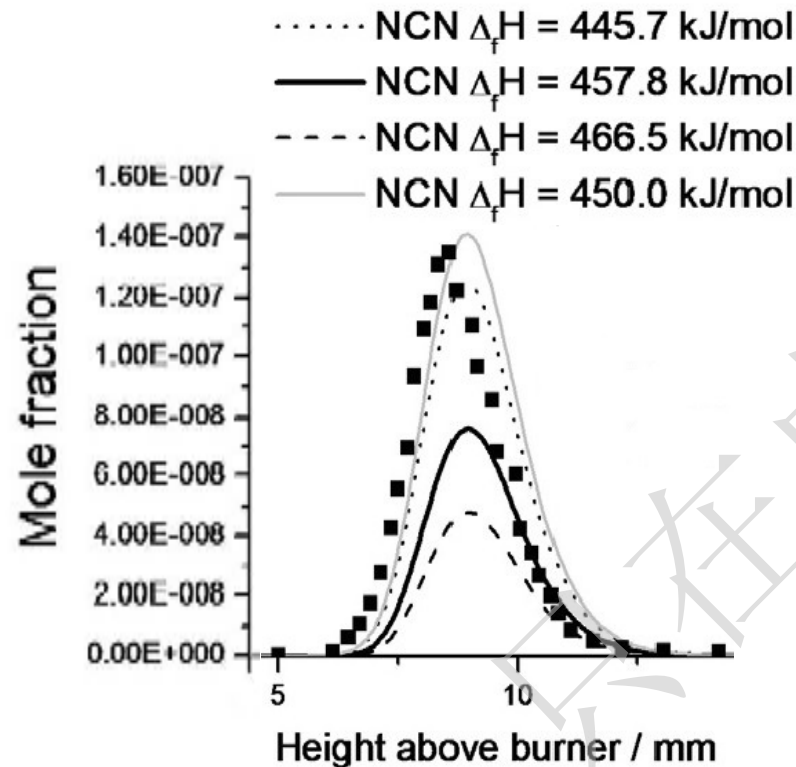
$$U = Nk_B T^2 \left(\frac{\partial \ln Q}{\partial T} \right)_V$$

- n Internal energy to calculate heat capability

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

1.2. CT in Combustion Chemistry

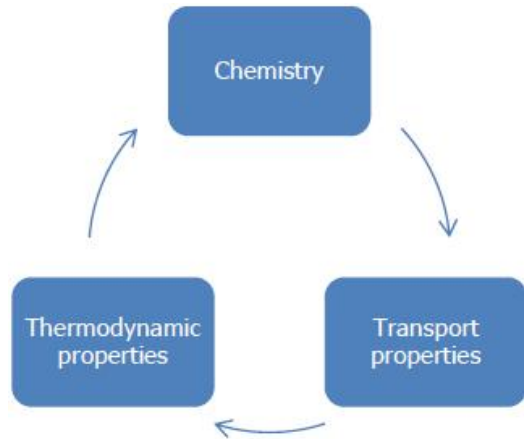
I Influence on Kinetic Modeling



NCN in a rich $\text{CH}_4/\text{O}_2/\text{N}_2$ flame at 40 Torr

$\Delta_f H^\circ_{298\text{K}}(\text{NCN})$, kJ/mol	Experimental or calculated values
501 ± 25	Gurvich [32]
473 ± 21	JANAF [33], listed by NIST web book [35]
465 ± 8	Martin et al. [36] multireference methods (recalculated from atomization energy)
466.5 ± 2.9	Bise et al. [11] (recalculated from $\Delta H_f^\circ_{0\text{K}} = 466.0 \pm 2.9$ kJ/mol)
464	Moskaleva and Lin [4], Mebel et al. [48]
Unknown value	
459	Harding et al. [5] multireference methods (calculated from reaction endothermicity of $\text{CH} + \text{N} \rightarrow \text{NCN} + \text{H}$)
457.8 ± 2.0	This work, recommended value
451.8 ± 16.7	Clifford et al. [10]
450	Melius [54]
448.7 ± 3.4	Canneaux et al. [12], weighted average of single reference methods
444.5	Dammeier and Friedrichs [57] (calculated from $\Delta H_f^\circ_{0\text{K}} = 444$ kJ/mol)
443–452	This work, single reference methods

2. Transport Properties



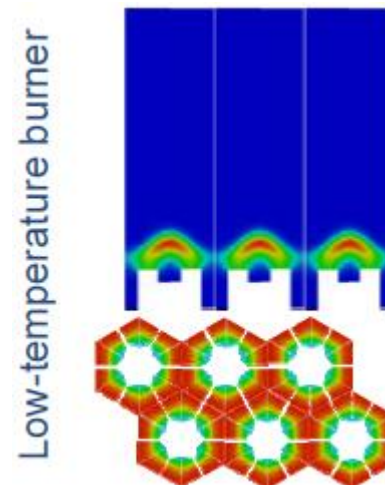
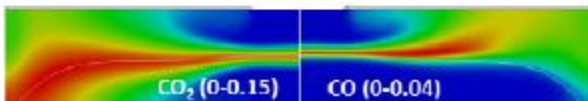
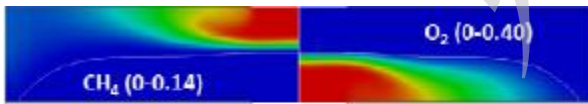
n Just as chemical reactions and their underlying parameters are essential for combustion modeling, molecular transport is important as well for non-homogeneous combustion system

n Viscosity

n Binary diffusion coefficients for combustion species

n Thermal conductivity

n Thermal diffusion



2. Transport Properties

I Transport Equations

Conservation Law

Continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla(\rho \mathbf{v}) = 0$$

Momentum equation

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \rho \mathbf{v} \nabla \mathbf{v} = -\nabla p - [\nabla \cdot \boldsymbol{\tau}] + \rho \mathbf{g}$$

Viscous tensor (fluxes of momentum)

Energy equation

$$\rho C_P \frac{\partial T}{\partial t} + \rho C_P \mathbf{v} \nabla T = -\nabla \mathbf{q} - \left. \frac{\partial \ln \rho}{\partial \ln T} \right|_p \frac{Dp}{Dt} - \boldsymbol{\tau} : \nabla \mathbf{v}$$

Heat flux

Mass flux

Viscous dissipation term (usually negligible)

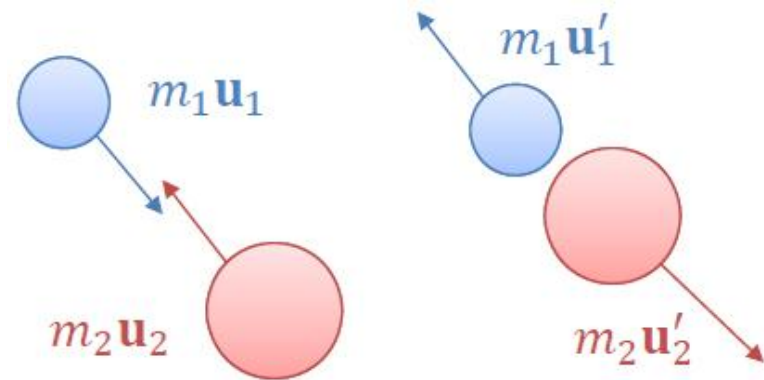
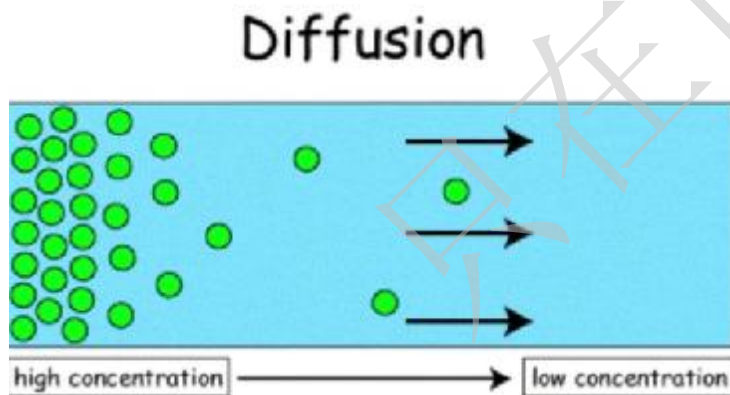
Species equations

$$\rho \frac{\partial \omega_k}{\partial t} + \rho \mathbf{v} \nabla \omega_k = -\nabla \mathbf{j}_k + \dot{\Omega}_k$$

2. Transport Properties

I Simple Theory

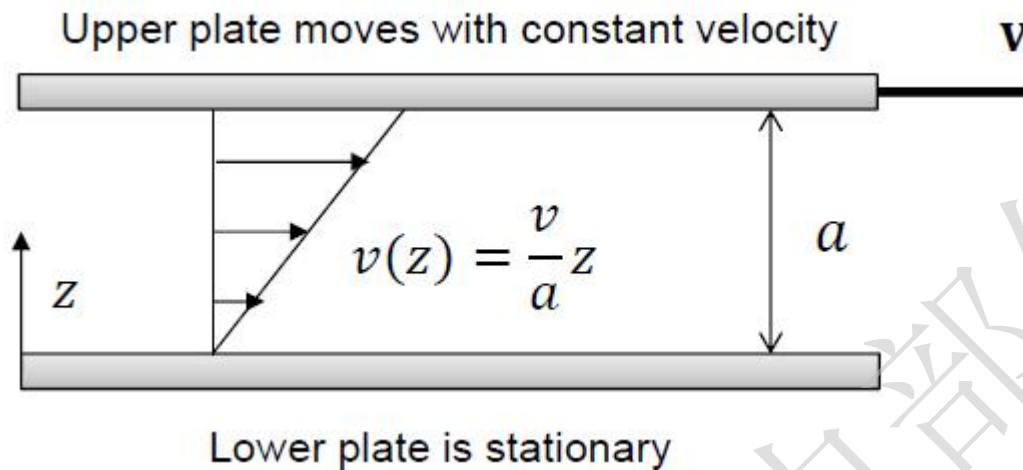
	Velocity gradient	Temperature gradient	Concentration gradient
Momentum	Newton's law		
Energy		Fourier's law	Dufour effect
Mass		Soret effect	Fick's law



Heat and mass transfer

2. Transport Properties

I Simple Theory – Viscosity



Newton's Law

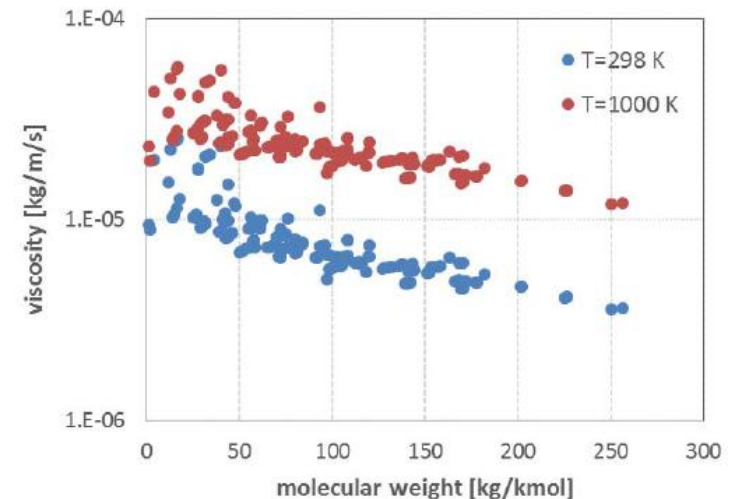
Momentum flux

$$f = -\mu \frac{dv}{dz} = \mu \frac{v}{a}$$

Experimental observations

- Increases with temperature as $\sim T^{1/2}$
- Almost independent of pressure
- Decreases with molecular weight

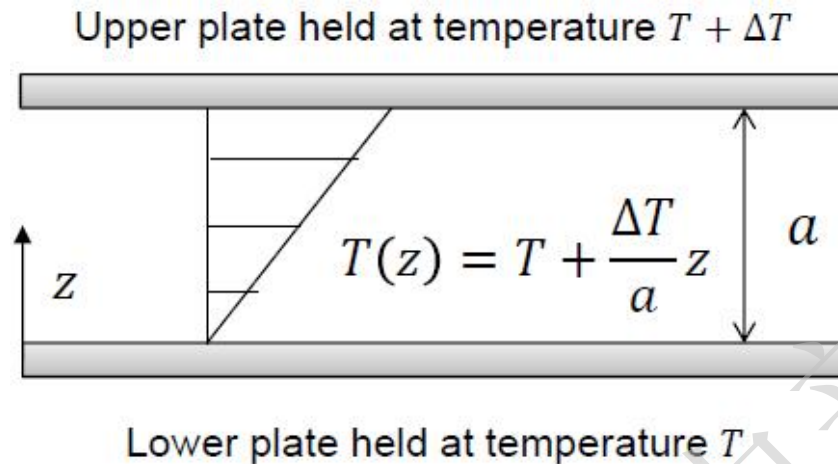
Viscosities of pure species in gas phase



2. Transport Properties

I Simple Theory – Thermal Conductivity

Fourier's Law



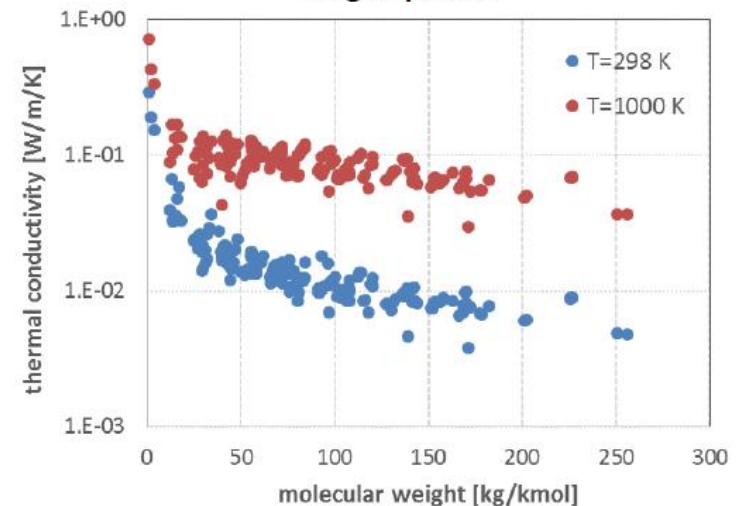
Heat flux

$$q = -\lambda \frac{dT}{dz} = -\lambda \frac{\Delta T}{a}$$

Experimental observations

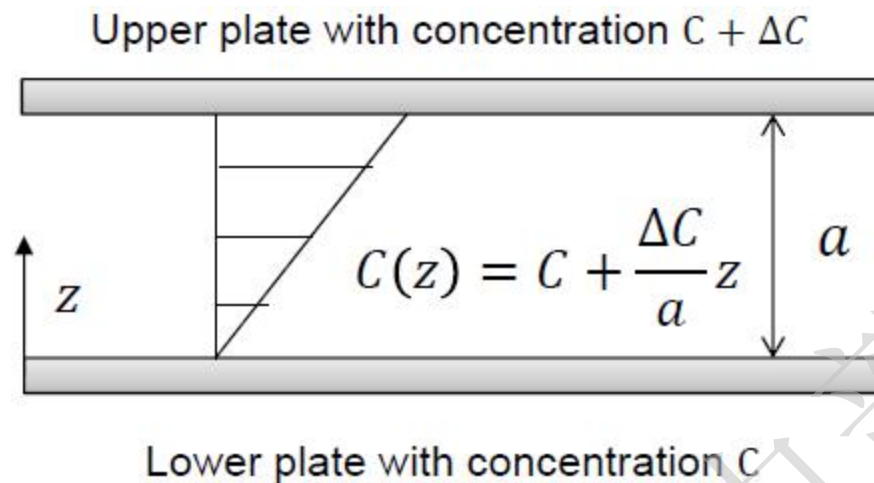
- Increases with temperature as $\sim T^{1/2}$
- Almost independent of pressure
- Decreases with molecular weight

Thermal conductivities of pure species in gas phase



2. Transport Properties

I Simple Theory – Mass Diffusivity



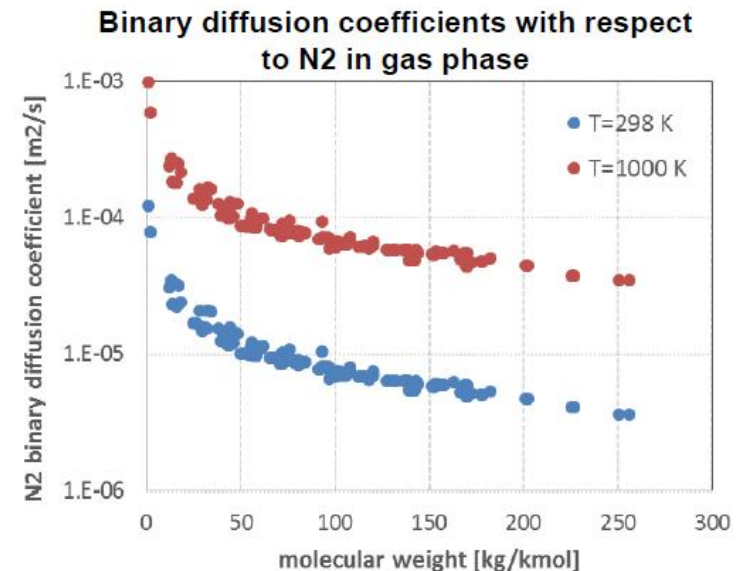
Fick's Law

Molar flux

$$J^* = -D \frac{dC}{dz} = -D \frac{\Delta C}{a}$$

Experimental observations

- Increases with temperature as $\sim T^{3/2}$
- Decreases with pressure as $1/p$
- Decreases with molecular weight



2. Transport Properties

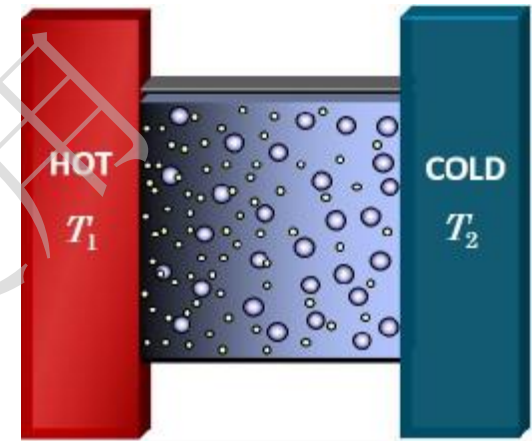
I Simple Theory – Thermal Diffusion Coefficients

Soret effect (thermal diffusion)

$$J^{th} = -\frac{D^T}{T} \nabla T = -\overset{\text{Fick's diffusion coefficient}}{D} \overset{\text{Thermal diffusion ratio}}{\theta} \frac{1}{T} \nabla T$$

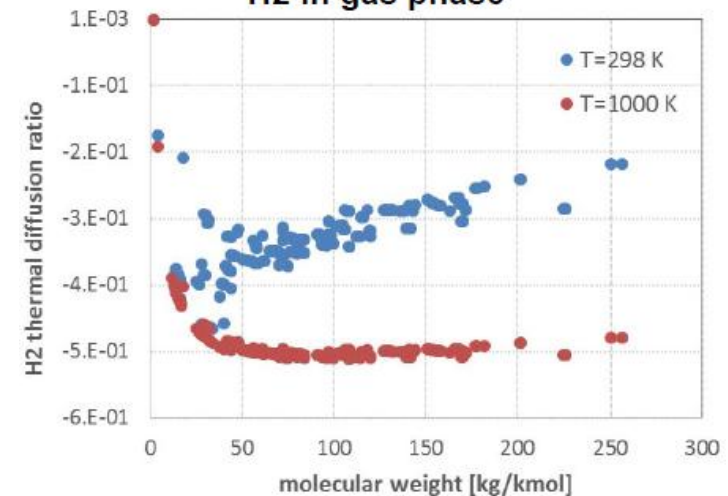
Fick's diffusion coefficient

Thermal diffusion ratio



- Thermal diffusion is important for very small species such as H_2 , H and He
- Need great temperature gradient to be important

Thermal diffusion ratio with respect to H_2 in gas phase



2. Transport Properties

I Kinetic Theory of Gases

Number of
molecules per unit
volume

$$n_V = \frac{N}{V} = \frac{p}{k_B T}$$

Mean free path

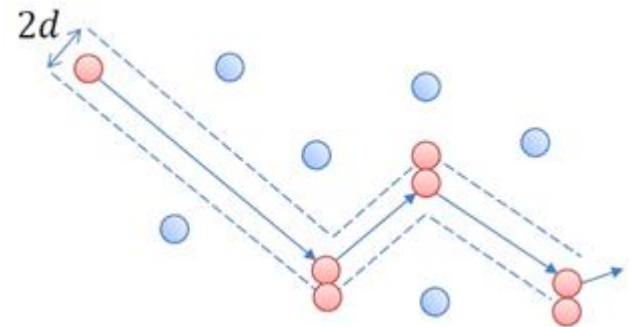
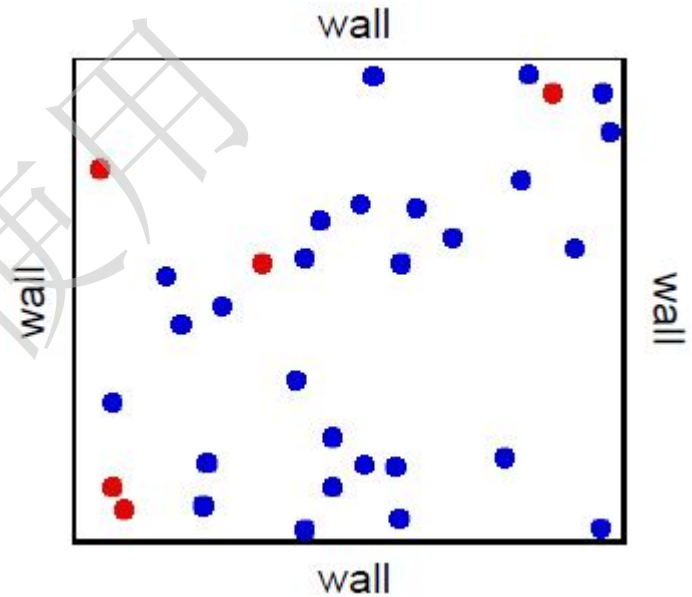
$$l = \frac{1}{\sqrt{2} \pi d^2 n_V}$$

Mean speed

$$\bar{c} = \sqrt{\frac{8k_B T}{\pi m}}$$

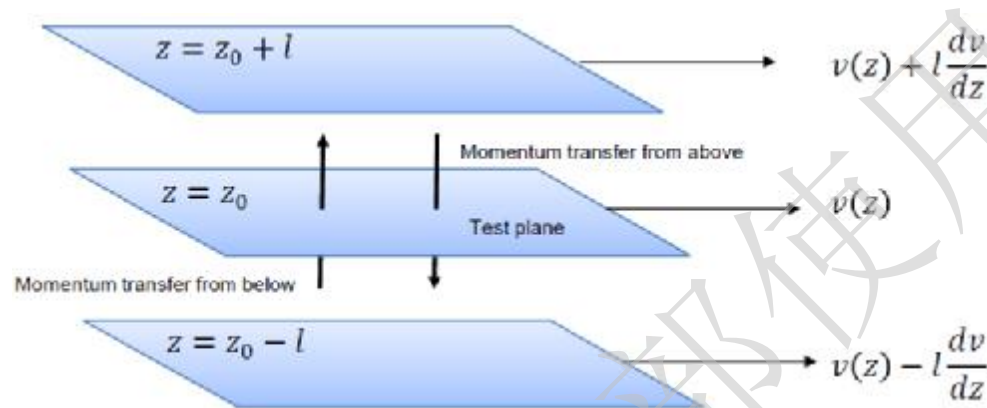
Number of
molecules colliding
with the wall per unit
area and unit time

$$Z = \frac{\bar{c}}{4} n_V = n_V \sqrt{\frac{k_B T}{2\pi m}}$$



2. Transport Properties

I Kinetic Theory of Gases – Viscosity



$$n_V = \frac{N}{V} = \frac{p}{k_B T}$$

$$l = \frac{1}{\sqrt{2} \pi d^2 n_V}$$

$$\bar{c} = \sqrt{\frac{8 k_B T}{\pi m}}$$

$$Z = \frac{\bar{c}}{\lambda} n_V = n_V \sqrt{\frac{k_B T}{2 \pi m}}$$

Net time rate of momentum change:

$$M = Zm \left\{ \left[v(z) - l \left(\frac{du}{dz} \right) \right] - \left[v(z) + l \left(\frac{du}{dz} \right) \right] \right\} = -2Zml \frac{dv}{dz}$$

$$\begin{cases} f = M = -2Zml \frac{du}{dz} \\ f = -\mu \frac{du}{dz} \end{cases}$$

Newton's Law



$$\mu = 2Zml = \frac{\sqrt{\pi m k_B T}}{\pi^2 d^2}$$

n Independent of pressure

n Increases with $T^{1/2}$

2. Transport Properties

I Kinetic Theory of Gases – Thermal Conductivity

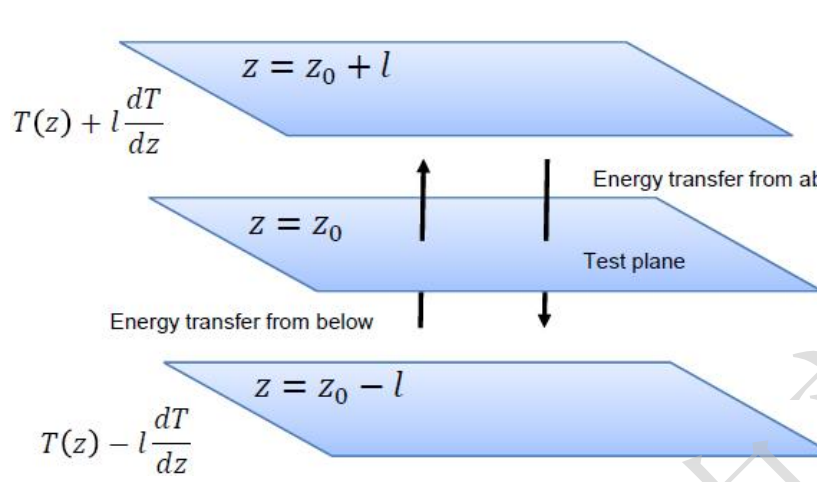


Diagram illustrating the kinetic theory of gases for thermal conductivity. A central "Test plane" is at $z = z_0$. Two other planes are at $z = z_0 + l$ and $z = z_0 - l$. Arrows indicate "Energy transfer from above" and "Energy transfer from below". The temperature at the upper plane is $T(z) + l \frac{dT}{dz}$ and at the lower plane is $T(z) - l \frac{dT}{dz}$. The energy per molecule at the central plane is $\varepsilon(z)$.

$$\varepsilon(z+l) = \varepsilon(z) + l \frac{C_V}{N_{AV}} \frac{dT}{dz}$$

$$\varepsilon(z-l) = \varepsilon(z) - l \frac{C_V}{N_{AV}} \frac{dT}{dz}$$

$$n_V = \frac{N}{V} = \frac{p}{k_B T}$$

$$l = \frac{1}{\sqrt{2} \pi d^2 n_V}$$

$$\bar{c} = \sqrt{\frac{8 k_B T}{\pi m}}$$

$$\lambda = \frac{\bar{c}}{4} n_V = n_V \sqrt{\frac{k_B T}{2 \pi m}}$$

Net heat flux:

$$q = Z \left\{ \left[\varepsilon(z) - l \frac{C_V}{N_{AV}} \left(\frac{dT}{dz} \right) \right] - \left[\varepsilon(z) + l \frac{C_V}{N_{AV}} \left(\frac{dT}{dz} \right) \right] \right\} = -2Zl \frac{C_V}{N_{AV}} \frac{dT}{dz}$$

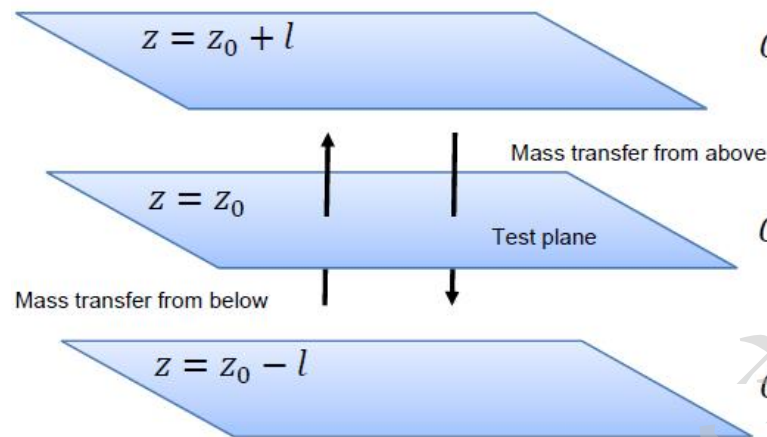
$$\left\{ \begin{array}{l} q = -2Zl \frac{C_V}{N_{AV}} \frac{dT}{dz} = \frac{\bar{c} n_V}{2 N_{AV}} l C_V \frac{dT}{dz} \\ q = -\lambda \frac{dT}{dz} \end{array} \right. \Rightarrow \lambda = \frac{\bar{c} n_V}{2 N_{AV}} l C_V = \frac{1}{\pi \sqrt{\pi}} \sqrt{\frac{k_B T}{m}} \frac{C_V}{N_{AV} d^2}$$

Fourier's Law

- n Independent of pressure
- n Increases with $T^{1/2}$

2. Transport Properties

I Kinetic Theory of Gases – Mass Diffusivity



$$C(z + l) = C(z) + l \frac{dC}{dz}$$

$C(z)$ Concentration of species

$$C(z - l) = C(z) - l \frac{dC}{dz}$$

$$n_v = \frac{N}{V} = \frac{p}{k_B T}$$

$$l = \frac{1}{\sqrt{2} \pi d^2 n_v}$$

$$\bar{c} = \sqrt{\frac{8k_B T}{\pi m}}$$

$$\lambda = \frac{\bar{c}}{n_v} = n_v \sqrt{\frac{k_B T}{2\pi m}}$$

Net molecular flux:

$$J^* = \frac{Z}{n_v} \left\{ \left[C(z) - l \left(\frac{dC}{dz} \right) \right] - \left[C(z) + l \left(\frac{dC}{dz} \right) \right] \right\} = -2 \frac{Z}{n_v} l \frac{dC}{dz}$$

$$\left\{ \begin{array}{l} J^* = 2 \frac{Z}{n_v} l \frac{dC}{dz} = \frac{\bar{c}}{2} l \frac{dC}{dz} \\ J^* = -D \frac{dC}{dz} \end{array} \right.$$

Fick's Law



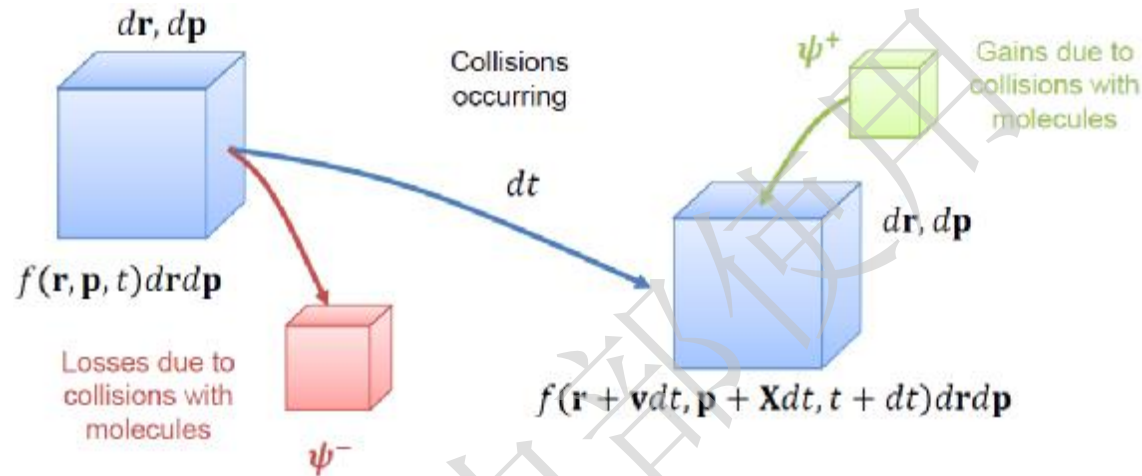
$$D = \frac{\bar{c}}{2} l = \frac{1}{\pi \sqrt{\pi}} \sqrt{\frac{k_B^3 T^3}{m}} \frac{1}{p d^2}$$

n Decreases with pressure as p^{-1}

n Increases with the temperature as $T^{3/2}$

2. Transport Properties

I Rigorous Theory – Boltzmann Equation

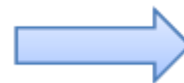


- n The Boltzmann equation is an integro-differential equation describing the time evolution of t

**Boltzmann
Equation**

$$\frac{\partial f}{\partial t} + \frac{\mathbf{p}}{m} \frac{\partial f}{\partial \mathbf{r}} + \mathbf{X} \frac{\partial f}{\partial \mathbf{p}} = [\psi^+ - \psi^-]$$

The loss and gain terms contain contributions from all possible collisions that can deflect the molecules during the time step dt



They are described through proper collision integrals

2. Transport Properties

I Rigorous Theory – Boltzmann Equation

From molecular distribution

$$\left\{ \begin{array}{l} \mathbf{q} = \int f \frac{mV^2}{2} V_i d\mathbf{v} \\ \mathbf{q} = -\lambda \nabla T \end{array} \right.$$

Fourier's law

Chapman-Enskog solution
of the Boltzmann equation



$$\lambda = \frac{75}{64} \left(\frac{k_B^3 T \pi}{m} \right)^{1/2} \frac{1}{\Omega^{(2,2)}}$$

From molecular distribution

$$\left\{ \begin{array}{l} T_{ij} = \int f m \mathbf{V} \mathbf{V} d\mathbf{v} \\ T_{ij} = -\mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) \end{array} \right.$$

Newton's law

Chapman-Enskog solution
of the Boltzmann equation



$$\mu = \frac{5}{16} (m \pi k_B T)^{1/2} \frac{1}{\Omega^{(2,2)}}$$

Collision integrals Ω :
the contributions from
all possible collisions
can deflect molecules.
Very important and
needs to be calculated
in kinetic modeling.

Reduced relative velocity

$$\Omega_{ij}^{(l,s)} = \left(\frac{2\pi k_B T}{m_{ij}} \right)^{1/2} \int \int e^{-\gamma_{ij}^2} \gamma_{ij}^{2s+3} (1 - \cos^l \chi) b db d\gamma_{ij}$$

Intermolecular potential

deflection angle $\chi(b, \gamma_{ij}) = \pi - 2b \int_{r_m}^{\infty} \frac{1}{r^2} \left(1 - \frac{b^2}{r^2} - \frac{u_{ij}(r)}{k_B T \gamma_{ij}^2} \right)^{-1/2} dr$

2. Transport Properties

I Rigorous Theory – Intermolecular Potential

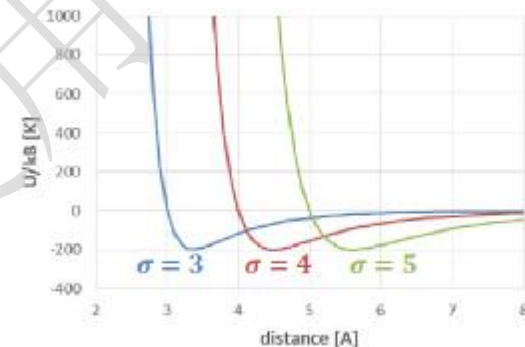
Lennard-jones potential (For monoatomic molecules)

$$U(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

n ε : net collision diameter

n σ : well-depth

Species	ε/k_B (K)	σ (Å)
Ar	136.5	3.330
O2	107.4	3.458
N2	97.53	3.621
CO2	244.0	3.763
CH4	141.4	3.746
H2O	572.4	2.605



Stockmayer potential (For diatomic and polyatomic molecules)

$$U(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 - \delta^* \left(\frac{\sigma}{r} \right)^3 \right]$$

n The additional term accounts for the interaction arising from permanent, non-uniform electric densities within each molecules

$$\delta^* = \frac{1}{4} \frac{\bar{\mu} \bar{\mu}}{\varepsilon \sigma^3} \chi$$

$\bar{\mu}$ Dipole moment

n Geometric function depending on the orientation of the molecules

n In the case where one or both the dipole moments are zero, the Stockmayer potential can be reduced to the Lennard-Jones potential

2. Transport Properties

I Rigorous Theory – Reduced Collision Integrals

Lennard-Jones potential

$$\begin{cases} \Omega^{(1,1)*}(T^*) \approx a_1 T^{*-a_2} + (T^* + a_3)^{-a_4} \\ \Omega^{(2,2)*}(T^*) \approx b_1 T^{*-b_2} + (T^* + b_3)^{-b_4} \end{cases}$$

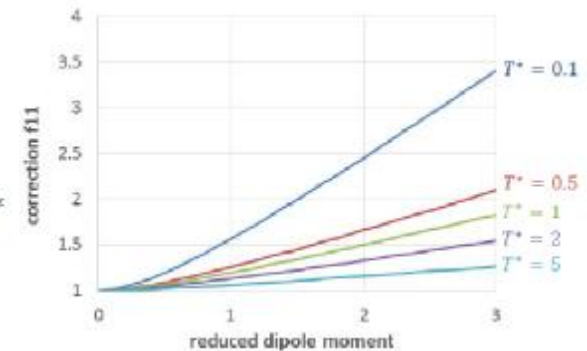
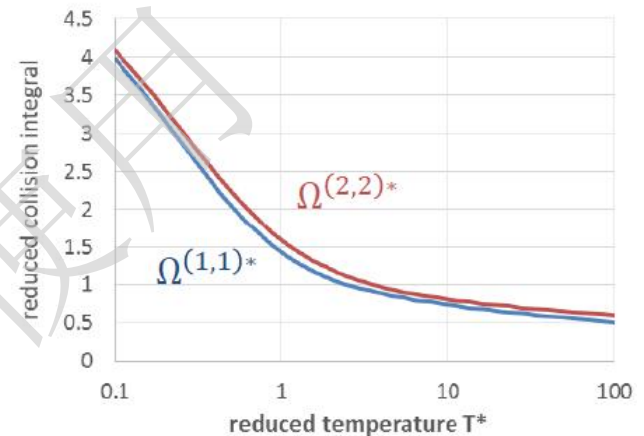
Stockmayer potential

$$\begin{cases} \Omega^{(1,1)*}(T^*) \approx [a_1 T^{*-a_2} + (T^* + a_3)^{-a_4}] f^{(1,1)} \\ \Omega^{(2,2)*}(T^*) \approx [b_1 T^{*-b_2} + (T^* + b_3)^{-b_4}] f^{(2,2)} \end{cases}$$

$$\begin{cases} f^{(1,1)}(T^*, \tilde{\delta}^*) = 1 + \frac{(e^{a_5/T^*} - e^{-a_6/T^*})(\tilde{\delta}^*)^2}{2 + 2.5\tilde{\delta}^*} \\ f^{(2,2)}(T^*, \tilde{\delta}^*) = 1 + \frac{(e^{b_5/T^*} - e^{-b_6/T^*})(\tilde{\delta}^*)^2}{2 + 2.5\tilde{\delta}^*} \end{cases}$$

$$\tilde{\delta}^* = 2\delta^*$$

Correction factors



In real gases the cross-section depends on the temperature. The reduced collision integral accounts for these dependencies.

2. Transport Properties

I Theory Summary

	Simple theory	Rigorous theory (Hard spheres)	Rigorous theory (general case)
Viscosity	$\mu = \frac{1}{\pi} \frac{\sqrt{\pi m k_B T}}{\pi \sigma^2}$	$\mu = \frac{5}{16} \frac{\sqrt{\pi m k_B T}}{\pi \sigma^2}$	$\mu = \frac{5}{16} \frac{\sqrt{\pi m k_B T}}{\pi \sigma^2} \frac{1}{\Omega^{(2,2)*}}$
Thermal conductivity	$\lambda = \frac{1}{\pi} \sqrt{\frac{k_B T}{\pi m}} \frac{C_v}{N_{AV} \sigma^2}$	$\lambda = \frac{25}{32} \sqrt{\frac{k_B T}{\pi m}} \frac{C_v}{N_{AV} \sigma^2}$	$\lambda = \frac{25}{32} \sqrt{\frac{k_B T}{\pi m}} \frac{C_v}{N_{AV} \sigma^2} \frac{1}{\Omega^{(2,2)*}}$
Mass diffusivity	$\Gamma = \frac{1}{\pi} \sqrt{\frac{\pi k_B^3 T^3}{m}} \frac{1}{\pi p \sigma^2}$	$\Gamma = \frac{3}{8} \sqrt{\frac{\pi k_B^3 T^3}{m}} \frac{1}{\pi p \sigma^2}$	$\Gamma = \frac{3}{8} \sqrt{\frac{\pi k_B^3 T^3}{m}} \frac{1}{\pi p \sigma^2} \frac{1}{\Omega^{(1,1)*}}$

- n The rigorous theory for hard sphere molecules gives the same general form of the transport properties evaluated through the simple model (i.e. the dependencies of T, p and m are exactly the same). Only the numerical coefficients are different (because of non-rigorous assumptions in the simple model).

2. Transport Properties

I Mass Diffusion – Binary Diffusion Coefficients

Convective and diffusive velocities

Species equations

$$\rho \frac{\partial \omega_k}{\partial t} + \rho \mathbf{v} \nabla \omega_k = -\nabla \cdot \mathbf{j}_k + \dot{\Omega}_k$$

Mass flux 

Conservation of momentum after collision

$$m_1 \mathbf{u}_1 + m_2 \mathbf{u}_2 = m_1 \mathbf{u}'_1 + m_2 m_1 \mathbf{u}'_2$$

Stefan-Maxwell equation for binary mixtures and Fick's law for binary mixtures

$$\nabla X_k = \sum_{j=1}^{NC} \frac{X_k X_j}{\Gamma_{kj}} (\mathbf{v}_j - \mathbf{v}_k) + (\omega_k - X_k) \frac{\nabla p}{p} + \sum_{j=1}^{NC} \frac{X_k X_j}{\rho \Gamma_{kj}} \left(\frac{\theta_j}{\omega_j} - \frac{\theta_k}{\omega_k} \right) \frac{\nabla T}{T}$$

The constraint equation:

$$\sum_{k=1}^{NC} \rho \omega_k \mathbf{v}_k = 0$$

Binary diffusion coefficients:

$$\Gamma_{jk} = \frac{3}{16} \sqrt{\frac{\pi k_B^3 T^3}{m_{jk}}} \frac{1}{p \pi \sigma_{jk}^2 \Omega_{jk}^{(1,1)*}}$$

2. Transport Properties

I Mass Diffusion – Mixture-Averaged

- n Approximate methods to determine transport properties in a mixture from the pure species values and binary diffusion coefficients
- n Compromise between accuracy and computational expense

$$\mu = \sum_{k=1}^K \frac{X_k \mu_k}{\sum_{j=1}^K X_j \Phi_{kj}}$$

The Wilke formula for mixture viscosity

$$\Phi_{kj} = \frac{1}{\sqrt{8}} \left(1 + \frac{W_k}{W_j} \right)^{-1/2} \left(1 + \left(\frac{\mu_k}{\mu_j} \right)^{1/2} \left(\frac{W_j}{W_k} \right)^{1/4} \right)^2$$

μ_k : the pure species k viscosity

$$\lambda = \frac{1}{2} \left(\sum_{k=1}^K X_k \lambda_k + \frac{1}{\sum_{k=1}^K X_k / \lambda_k} \right)$$

Mixture-averaged thermal conductivity

$$D'_{km} = \frac{1 - Y_k}{\sum_{j \neq k}^K X_j / \mathcal{D}_{kj}}$$

Mixture-averaged diffusion coefficients

- n No reliable mixture-averaged theory is available for computing the thermal diffusion coefficients

2. Transport Properties

I Mass Diffusion – Multicomponent

- The Chapman-Enskog theory provides the basis for multicomponent transport properties. The thermal conductivities and the thermal diffusion coefficients can be calculated from the solution of a system of linear equations defined by the L matrix

Nine-block submatrices

$$\begin{bmatrix} \mathbf{L}^{00,00} & \mathbf{L}^{00,10} & \mathbf{0} \\ \mathbf{L}^{10,00} & \mathbf{L}^{10,10} & \mathbf{L}^{10,01} \\ \mathbf{0} & \mathbf{L}^{01,10} & \mathbf{L}^{01,01} \end{bmatrix} \begin{bmatrix} \mathbf{a}_{00}^1 \\ \mathbf{a}_{10}^1 \\ \mathbf{a}_{01}^1 \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{X} \\ \mathbf{X} \end{bmatrix}$$

Vector of mole fractions

Thermal conductivity

Thermal coefficients

Thermal diffusion ratios

Thermal diffusion velocity

$$\lambda = \lambda_{trans} + \lambda_{int}$$

$$\lambda_{trans} = -4 \sum_{k=1}^{NC} X_k a_{10,k}^1$$

$$\lambda_{int} = -4 \sum_{k=1}^{NC} X_k a_{01,k}^1$$

$$\theta_k = \frac{8m_k X_k}{5R} a_{00,k}^1$$

$$\theta_k = \sum_{j \neq k}^{NC} \theta_{kj}$$

$$\mathbf{v}_k^T = -\frac{D_{k,mix} \theta_k}{X_k} \frac{1}{T} \nabla T$$

$$\theta_{jk} = \frac{15}{2} \frac{(2A_{jk}^* + 5)(6C_{jk}^* - 5)}{A_{jk}^*(16A_{jk}^* - 12B_{jk}^* + 55)} \frac{MW_j - MW_k}{MW_j + MW_k} X_j X_k$$

- When thermal diffusion is important, the rigorous multicomponent theory should be used

2. Transport Properties

I Transport Data for Kinetic Modeling

Example of file containing the transport data for each species in TRANLIB format

Species	Geometry	ϵ/k_B	σ	μ	α	Z_{rot}
H2	1	38.000	2.920	0.000	0.790	280.000
HE	0	10.200	2.576	0.000	0.000	0.000
C6H6	2	468.500	5.230	0.000	10.30	1.000
H2O	2	572.400	2.605	1.844	0.000	4.000

Name of the species
(must be the same
reported in the
thermodynamic and
kinetic files)

The potential well
depth (in K).

The collision diameter
(in angstroms)

The dipole moment
(in Debye)

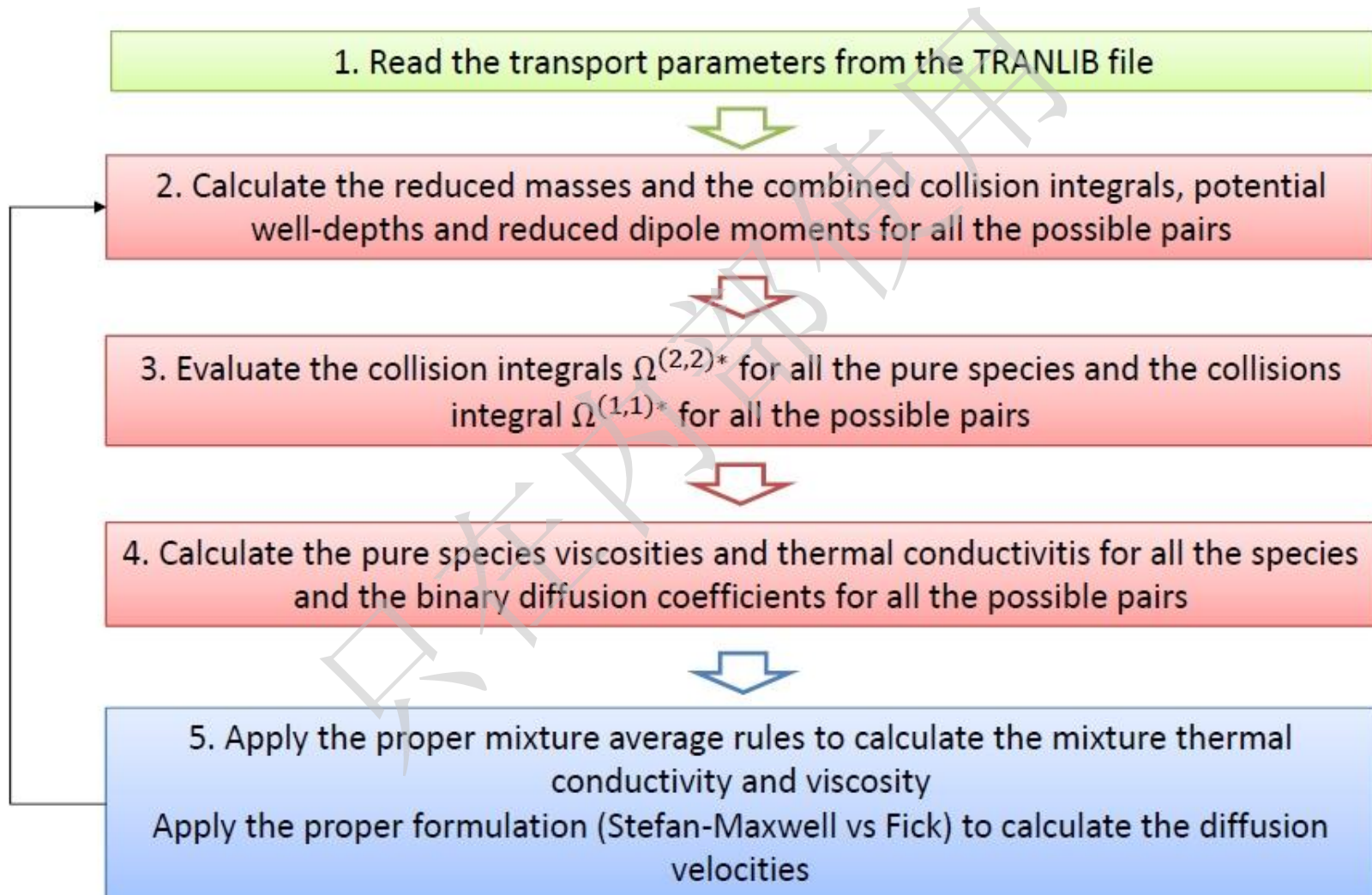
The polarizability (in
cubic angstroms)

The rotational relaxation
collision number at 298 K

An index indicating the geometrical
configuration. If the index is 0, the
molecule is a single atom. If the index is
1 it is linear, and if it is 2, the molecule
is nonlinear.

2. Transport Properties

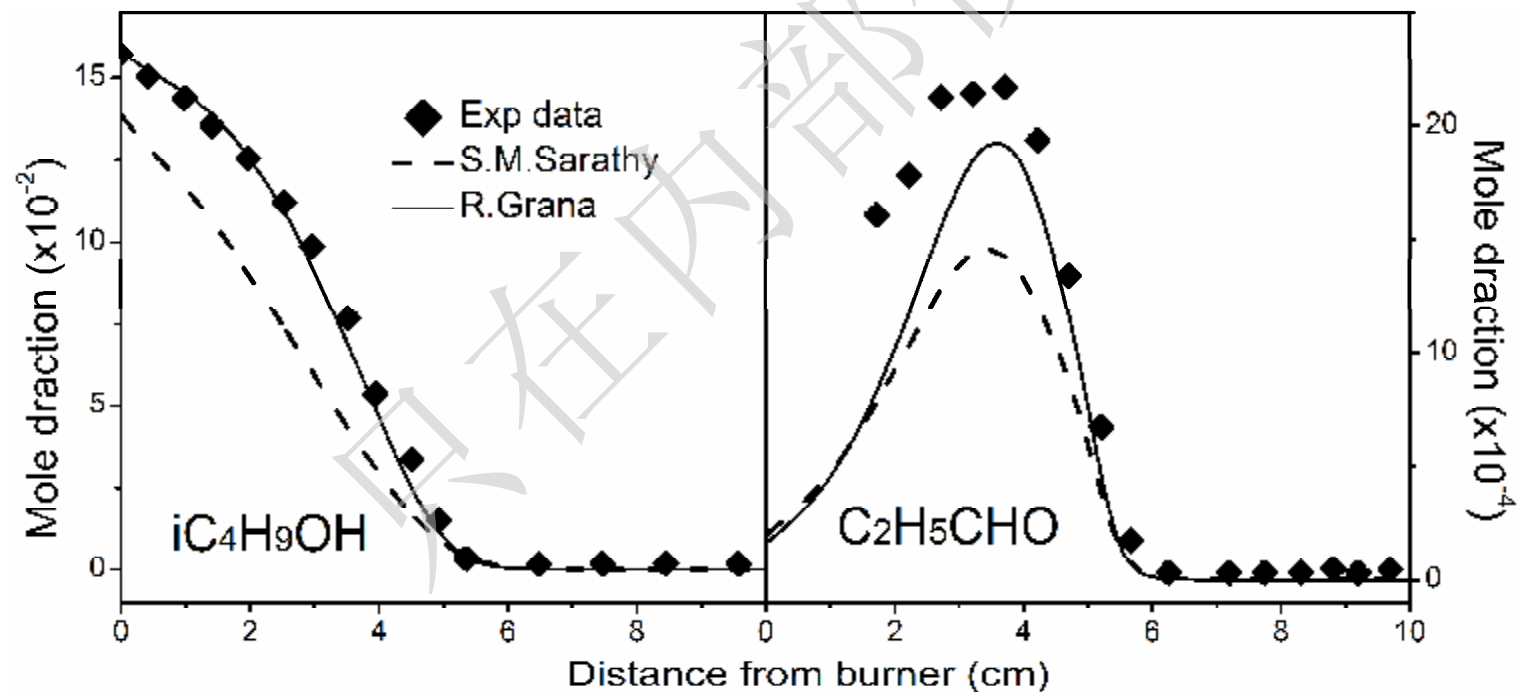
I Calculation of Transport Properties



2. Transport Properties

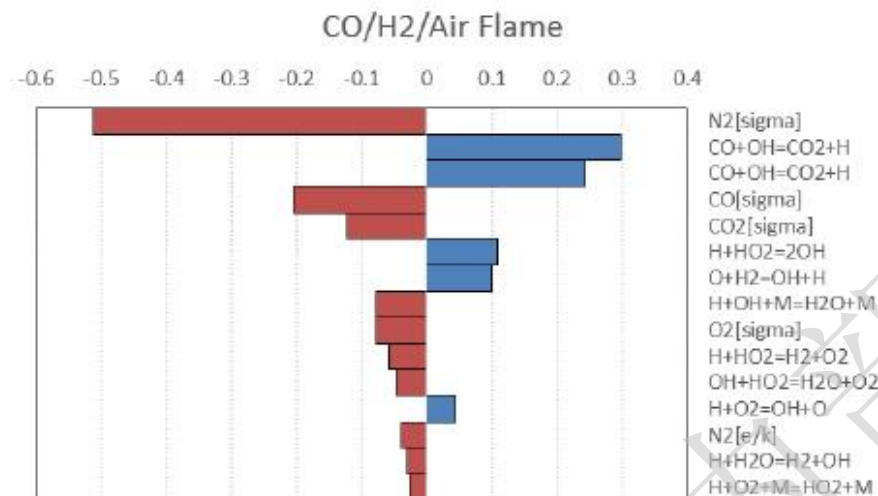
I Importance for flame simulation

Species Name	Geometry	ε/k_B	σ	μ	α	$Z_{\text{rot.}}$
iC4H9OH (S.M.Sarathy)	2	641.92	2.40	1.67	8.88	1.0
iC4H9OH (R.Granda)	2	503.07	5.33	1.67	8.88	1.0



2. Transport Properties

I Importance for flame simulation



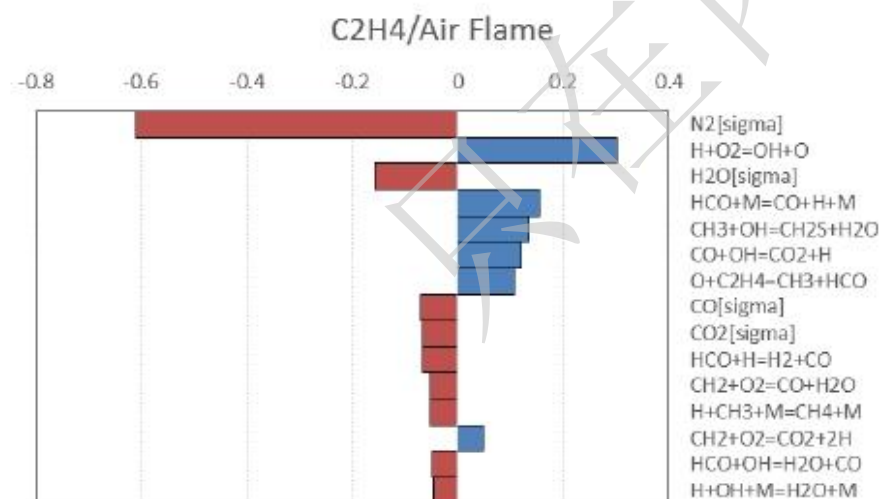
Significant sensitivity to the collision diameter σ of:

N₂, O₂, CO, H₂O, CO₂

Expected since transport properties depends on the squared collision diameter $\sim 1/\sigma^2$

Smaller sensitivity to the well depths ε of:

H, N₂, CO, H₂O, CO₂



The smaller sensitivity to ε is reasonable because ε is used as scaling factor for the definition of reduced collision integral (through the reduced temperature)

2. Transport Properties

I Recent Progresses



Contents lists available at [ScienceDirect](#)

Progress in Energy and Combustion Science

journal homepage: www.elsevier.com/locate/pecs



Contents lists available at [ScienceDirect](#)

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

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First-principles binary diffusion coefficients for H, H₂, and four normal alkanes + N₂

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3. Recommended Literature

1. PART1, Equilibrium, *Atkins' Physical Chemistry*, 9th edition, edited by Peter Atkins and Julio de Paula, December 18, 2009. *For ones who want to know the detail about thermodynamics and statistical thermodynamics.*
2. Edward R. Ritter, Joseph W. Bozzelli, THERM: Thermodynamic property estimation for gas phase radicals and molecules. *International Journal of Chemical Kinetics*, 23 (1991) 767–778. *For ones who want to know more about Benson's group additivity method.*
3. Chapter 12, *Chemically Reacting Flow: theory and practice*, edited by Robert J. kee, Michael E. Coltrin and Peter Glarborg, 2003. *For ones who want to know more about transport phenomenon for combustion modeling.*
4. Nancy J. Brown, Lucas A.J. Bastien, Phillip N. Price, Transport properties for combustion modeling. *Progress in Energy and Combustion Science* 37 (2011) 565-582. *For ones who want to know more about transport phenomenon for combustion modeling.*

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